Landmark XVI-P — Scientific manuscript

Science volume (p23: O.1 requirement recorded, one-loop power counting, P.6 entry inventory (kappa^3 three-point, not evaluated); P.5 calculation renormalized at one loop kappa^2 for the two-point functions with QdS external legs; bulk symmetry not claimed; O.1 NOT_CLOSED)

XVI-P p23

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Science volume. Certificates and listings: `XVI-P-appendix.pdf`; combined: `XVI-P.pdf`.

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1 p23: the O.1 requirement, one-loop power counting and the P.6 entry inventory

O.1, exactly. The XVI-O ledger `O_quantum_target.json` (o70-sync; verbatim copy in `provenance/XVI-O`, sha pinned, read only) states the O.1 target: the one-loop κ^2 bulk two-point sector on S_{inv} (graviton, scalar, ghost, including the gh = 0 BV source seeds; frozen gauge fermion; $m = 0$). O.1 closes iff this sector closes under a *verified* retained quantum identity. Every checklist item must be earned: O.2 (classical κ^2 master, including grav ℓ_4), O.4 (inventory), O.5 (graphs evaluated), O.6 (identity verified) and O.7 (remainder cancellation / counterterm solution). The target must first be enlarged if the identity constrains more legs or orders at κ^2. *In XVI-P scope:* O.4, O.5, the projected part of O.6/O.7, and the enlarge rule. *Out of scope:* O.2 and the bulk ST4b identity, which belong to XVI-O/XVI-N and stay untouched. Hence O.1 is **not closable inside XVI-P**, and it stays NOT_CLOSED.

Exact one-loop power counting. Every interaction vertex of the frozen inventory carries $\kappa^{\text{legs}-2}$, with antifield sources counted as legs. This is checked vertex by vertex. A one-loop graph has $I = V$, so it is κ^E with E external legs. The one-loop identity at κ^2 is $(S^{(0)}, \Gamma_1^{(2)}) + (S^{(1)}, \Gamma_1^{(1)})$; the term $(S^{(2)}, \Gamma_1^{(0)})$ vanishes because $\Gamma_1^{(0)}$ is field independent. Only $E = 2$ and $E = 1$ enter. A mechanical enumeration of all one-loop 1PI graphs (declared propagators hh, $\phi\phi$, $c\bar{c}$) gives the following:

- for $E = 2$ it reproduces the frozen κ^2 inventory exactly (D01–D06, D11, D12, D13: 9/9);
- for $E = 1$ it gives the 3 graphs of the one-point row;
- it detects four mutations (a wrong κ power, two dropped vertices, a spurious propagator).

The enlarge rule is **not triggered**: there is no one-loop three-point sector at κ^2.

P.6 entry (honest partial). Three-point functions first appear at κ^3. The one-loop κ^3 three-point inventory has 15 graphs: hhh 5, $h\phi\phi$ 4, $c\bar{c}h$ 2, Kch 2, c^*cc 1, $\phi^*c\phi$ 1. This is a topology-level inventory only. The amplitude rules, the counterterm domain over $\mathbb{Q}(i)$ (including evanescent operators), the evaluation, the three-point defect and σ_0 closure are **not done** in p23, so no defect outcome is claimed.

Fixes from the p22 review. (i) Polynomial-domain refusal (p23 contract revision, one block appended to `dred.py`). From outside sympy, polynomial algebra that can evaluate d raises: `rem`, `prem`, `div`, `resultant`, `compose`, `Poly.shift`, `Poly.rem`, and similar. Internal renames of d are admitted only to a Dummy, so `Matrix.subs(d→y)` raises. A live probe runs 9 forms on 4 expressions, and rational-function algebra is unchanged. (ii) The gate now checks the values $\rho = 1/2$, E_2 coefficient $-1/16$, SE coefficient $+1/8$, and the summed h.h coefficients against both pinned values and the archived p22 certificate. (iii) The README title line carries "two-point functions with QdS external legs".

2 p22: evanescent components (EV-M2), evanescent counterterms (EV-CT) and the runtime d guard

Contract revision (disclosed). p22 carries one disclosed contract revision, listed in `p22_conventions.CONTRACT_REVISION`: (i) evanescent counterterms are admitted outside the p14r 4D domain, and (ii) `code/p5_entry/dred.py` gains one delimited additive block (the runtime d guard). Outside the block the file is byte-identical to the p18 release; `contract_registry.dred_revision` checks this, and the p22 gate pins the exact sha256 of the revised file.

EV-M2: the matching statement. XVI-O has no evanescent counterpart, so the eight evanescent ($\tilde{g}$) structures of D11 cannot be matched to XVI-O. We tested three candidate statements exactly, with d opaque: (a) *match a d-dimensional continuation of the XVI-O kernel* is refuted for both natural continuations ($g_4 \to \hat{g}$ and g_4 kept), because the exact D11 tensor differs from either continuation of its own 4D part on ˜external

legs. XVI-O defines no other continuation, so (a) is not a usable statement. (b) *Pure local evanescent operators*: proven. Every evanescent pole (D01, D03, D11) is a polynomial in p and independent of d after pole extraction. (c) *Vanishing in the renormalised result*: proven in a stronger form. After subtraction, every evanescent component is identically zero on 4D (hatted) external legs at every d, not merely at $O(\epsilon)$. The scope is `D11_matching_verified_scope` = "4D components: M2 discharged (P18_D11_M2, unchanged); evanescent components: EV-M2 proven; NOT matched to XVI-O".

EV-CT: the counterterms. The coefficient is −(UV pole) in the $N_\epsilon \kappa^2/\epsilon$ normalisation of P14. There are three pieces:

- h.h operators for the eight $\tilde{g}$ structures, summed over D01+D03+D11: $g_4\tilde{g}\,(p^2)^2$: $-7i/120$ (twice); $g_{4,mr}\tilde{g}_{ns}(p^2)^2$: $-i/160$; $\tilde{g}\tilde{g}\,(p^2)^2$: $i/160$ (twice); $\tilde{g}\,pp\,p^2$: $7i/120$ (twice), $i/160$. Per-graph values are in `certificates/P22_evanescent.json`.
- the K′.c operator for D13 E_2, with coefficient $-1/16$;
- its ghost-equation partner $\bar{c}\,\Sigma_{\rm SE}\,c$, with coefficient $+1/8$.

Closure. σ_0 of the counterterms is zero in every sector, so full closure holds with the evanescent counterterms included. In h.c, the per-graph $p \cdot T$ of D01 and D03 cancel in the sum, D11 is transverse, and $E \cdot E_2 = 0$. In c.b, the ghost equation with $\rho = 1/2$ (calibrated on the 4D pair) holds exactly in d for E_2 and $\Sigma_{\rm SE}$. c.cbs is unreachable. **Identity.** The h.h slot values of the counterterm sum are zero in all six external modes, and the K.c slot values are zero, so the projected identity is unchanged. **Remnants.** The finite evanescent parts are nonzero, but they vanish identically on hatted external legs. With QdS external legs there is no $O(\epsilon)\times$pole effect. If external traces are taken with g_4 (Q4S), a finite term $7i\,p^2(\hat{a}\cdot\hat{b}\,p^2 - \hat{a}\cdot p\,\hat{b}\cdot p)/60$ arises. It is disclosed as a scheme choice. **Mutations.** Each of the following is detected: dropping the D01 or D03 counterterm; doubling D03; dropping the SE partner; flipping the sign of the E_2 counterterm; and putting a trace operator in place of $s_0 h$. **Status.** P.5 calculation status is *renormalized* (one loop, κ^2, the two-point functions hh / K′.c / $\bar{c}$c, MS-bar in DRED with the p22 evanescent counterterms, external legs in QdS). It does not cover higher points, two loops, bulk symmetry or O.1.

Runtime d guard. The guard intercepts the following calls whenever the caller is outside sympy/mpmath/dred: `subs`, `xreplace` (including on a bare d), `replace`, `evalf/n`, the Poly value and coefficient accessors, `coeff`, `as_independent`, `as_coeff_exponent`, `series`, `collect`, `limit`, `Lambda`, and $d == d$. Any of them that would give d a value raises `PrematureD4Substitution`. Authorised paths (`substitute_d`, `authorised_d`) still work. A live probe in the gate runs 30 forms on 4 expressions: every reviewer form from p21, the power-base hole, alias and free_symbols lookups, `Symbol.__new__` and `Function` calls. All of them raise, and ordinary algebra is unchanged. Some bypasses cannot be intercepted at runtime: identity (`is`) tests, string round-trips and `Poly.rep`. These are disclosed and covered only by the secondary static allow-list scan.

Gate. The tools/ and notes prefix exemptions are removed. Every changed, added or removed file relative to the p21 zip must be listed in `DISCLOSED_REVISIONS_P22`. The only exceptions are the generated reference values and the pinned archival/p21 records. Earlier-layer gates accept the registered p22 revisions only through `contract_registry`.

3 Small release (p21): absolute D11 pin

Scope.

p21 is a small release from p20 with no scope expansion. It follows the findings of the p20 independent rebuild (PASS; report sha256 e8f358f9…, bound to tree 2c0aa153… and zip ca32ce01…; same box, not a second machine). No contract revision; the p15–p19 layers are as in p20; the disclosed p21 revisions of the p20 layer are `d11_pin.py` and `release_p20.py`. bulk_symmetry_restored is false regardless. O.1 stays NOT_CLOSED.

Absolute D11 pin (p20: OPEN). The sign and magnitude of the D11 UV pole are derived from $\frac{1}{2}\mathrm{Tr}\ln$ of the minimal scalar loop under the package's declared conventions, each checked verbatim in the code:

- $G = \Gamma^{(2)}$: the amplitude rules declare $i\Gamma_1 = \int \mathrm{pref}\, N/\prod(q^2 + i0)$ and G6 declares $G = \kappa^2 N_\epsilon \hat{G}$, $\hat{G} = -i\,\mathrm{pref}/\kappa^2 \int N/(D_1 D_2)$; hence $G = -i \cdot i\Gamma = \Gamma$.
- Wick direction: $N_\epsilon = iS_\epsilon/(16\pi^2)$ and the certified bubble pole $\hat{I}(1,1) = +1/\epsilon$ give $d^d l_M = +i\, d^d l_E$; with $\eta = (+,-,-,-)$ and $S_\phi = \frac{1}{2}\int \sqrt{-g}\, g^{\mu\nu}\partial\phi\partial\phi$ this gives $iS_M = -S_E$, so $L_M^{\mathrm{eff}} = -L_E^{\mathrm{eff}}$.
- Heat kernel (Euclidean, $d = 4 - 2\epsilon$): $L_{E,\mathrm{div}} = -a_2/(32\pi^2\epsilon)$, so $L_{M,\mathrm{div}} = +a_2/(32\pi^2\epsilon)$ with $a_2 = C^2/120 + R^2/72$ ($\xi = 0$).
- With $g = \eta + \kappa h$ the Minkowski quadratic forms are $C^2 \to \frac{1}{2}p^4 h P_2 h$ and $R^2 \to 3p^4 h P_0 h$.

Result: $\Gamma^{(2)}|_{\mathrm{pole}} = \kappa^2/(16\pi^2\epsilon)\,[p^4 P_2/240 + p^4 P_0/24]$. This equals the package pole $\kappa^2 N_\epsilon \hat{G}/\epsilon$ with $c_2 = -i/240$, $c_0 = -i/24$ exactly (ratio 1 for both spins). No convention choice is external; the external inputs are mathematics only (the a_2 coefficients and the heat-kernel pole formula). A $\times 2$ normalisation error and a sign flip keep the 10:1 ratio but are detected by the absolute pin.

Gate. The top-level `absolute_pin_status` and `ratio_pin_status` (the keys typeset in this PDF) must equal the nested statuses. Run standalone, the p20 gate re-checks the p18 layer (quantum inputs earned) and the graph inventory instead of trusting the P19 layer record; the p21 gate re-checks all lower layers.

Scan. The enumerated pattern list is complemented by an allow-list scan (`tests/test_p21.py`): expression-tree decomposition (`.args`, `make_args`, `as_ordered_terms`, ...), sympy callable aliases, computed symbol names and value calls mentioning d are refused everywhere except at reviewed sites. All eight forms reported by the reviewer are caught. A runtime check confirms that the stored exact-d D11 amplitudes re-parse with the opaque d and reproduce their poles only through `substitute_d`. It also found a limitation: `expr.subs(d, v)` is not intercepted when d occurs only as a power base. The contract is not changed in p21; no package value path uses it, and the static rule refuses it. This is a check, not a proof.

Disclosure. The p20 edit of `manuscript/notes/P19_defect.tex` is now listed in `DISCLOSED_MANUSCRIPT_REVISIONS`. The reference key `/p19/change_log_vs_p18`, which was recomputed on the current tree, is renamed `p19_gate_change_log_vs_p18_recomputed_on_current_tree`. The log as released stays in the pinned `archival/p19/reference_values_p19.json`.

4 Correction release (p20): erratum to p19

Scope.

p20 is a correction release of p19 with no scope expansion. It follows the findings of the p19 independent rebuild (PASS with findings; report sha256 1b01f1d8..., bound to tree fcf90354... and zip da924275...; same box, not a second machine). There is no contract revision, and the p15–p18 layers are byte-identical. The disclosed p20 revisions of the p19 layer are `defect.py` and `release_p19.py`. bulk_symmetry_restored is false regardless. O.1 stays NOT_CLOSED; the insertion stays frozen; XV-B and XIV-J are unopened; Route B is not discharged; no $c\,\delta$ is added to u_μ^R.

Erratum K1 (the p19 text is retracted). p19 stated that full σ_0 closure of the MS-bar counterterm fails with the derived ghost-equation factor $+1$, that it needs -1, and that this was unresolved. The statement is wrong; it came from a Grassmann-orientation bookkeeping error.

- The route-G densities are generated in the order (c,cb). The P15 ghost self-energy coordinates Σ use the (cb,c) orientation, and $V[\mathrm{c.cb}\#k] = -V[(\mathrm{cb,c})\#k]$. p20 measures this ratio (via `pw_vertex`) as -1 for $k = 0, 1$.

- In route-G (c,cb) coordinates, Σ has pole $(+3/8, -1/4)$ and local part $(2/3, -5/12)$.
- The derived factor is $+1$ (the ghost equation $\Sigma = 2iD[G_{Kc}]$ reproduces the P15 pole). With it, the *complete* MS-bar counterterm $S_{ct} = -$(h.h + c.K + c.cb poles) is σ_0-closed in every ghost $+1$ sector (h.c, c.b, c.cbs).
- Without the c.cb term, or with Σ unconverted (the p19 error), $\sigma_0(S_{ct})$ has 88 entries in the c.b sector.
- Tree-level controls: $V_{S_0}[\mathrm{cb}, \mathrm{c}] = 2iD\, V_{S_0}[K, c]$, and $\sigma_0(S_0^{Kc} + S_0^{\bar{c}c}) = 0$ (the S_0^{Kc} part alone has a nonzero image).
- In quotient coordinates, the c.K-only data have c.cb components $(-3/8, 1/4)$, which cancel against $\Sigma = (+3/8, -1/4)$. The full data including Σ have quotient coordinates 0.

Restoration redefined. The p19 flag (renormalised $P_{ch}\sigma_0(S_{ct}) = 0$, $x = 0$) follows from the identity and locality, so it is not independent. In p20, `projected_restoration_verified` means full σ_0 closure of the complete one-loop κ^2 MS-bar S_{ct} (including c.cb, Σ held fixed) in all ghost $+1$ sectors, together with the projected identity. Scope: that counterterm only; not bulk symmetry, not O.1. The corrected computation closes, so the flag is earned. The p19 check is kept as a labelled route-consistency control, and it holds.

Blind spots. The full-closure test acts as a mutation detector, with Σ held fixed:

- drop D13 and flip K.c are detected (c.b, 88 entries);
- drop D01, drop D03, flip D03 and D03 $\times 2$ are detected (h.c, 280 entries);
- the p19 orientation error is detected;
- drop D11 is not detected: D11 is transverse, so its pole is closed by itself.

D11 normalisation pin (`P20_D11_pin.json`).

- *Derived from package data.* The 4D UV pole of D11 (`P18_D11.json`, package orbit convention; transverse; only spin-2 and spin-0s parts) has Barnes–Rivers coefficients $c_2 = -i/240$ and $c_0 = -i/24$, so $c_0 : c_2 = 10$. The quadratic forms on $g = \eta + h$ are derived here from linearised curvature: $C^2 \to \frac{1}{2}p^4 h P_2 h$, $R^2 \to 3p^4 h P_0^s h$, and Gauss–Bonnet $\to 0$.
- *External literature.* The a_2 coefficient of a minimally coupled real scalar is $\frac{1}{120}C^2 + \frac{1}{72}R^2$ (Birrell–Davies §6.2; 't Hooft–Veltman 1974). This predicts $\frac{1}{240} : \frac{1}{24} = 1 : 10$.
- *Ratio pin: PASS.* Drop D11 gives $c_2 = c_0 = 0$ and is detected, and so is a swap of the two gg structures.
- *Absolute pin: OPEN* [**superseded in p21: the absolute pin was derived and verified, sign and magnitude (P21, section 1; certificates/P20_D11_pin.json, absolute_pin_status PASS); the text below is the p20 statement, kept for the record**]. The magnitudes agree up to a common factor $-i$, but the map from $\hat{G}$ to a_2 (phase, N_ϵ, κ normalisation, sign of Γ) is not derived. An overall $\times 2$ error is caught only by this open pin and by M2 (P18).

J1 wording. The $\mathbb{Q}(i)$ counts are the real dimensions under a compatible real structure. The earlier wording, "the physical field", is replaced.

Scan and gate. The static scan is still a pattern check, not a proof. It now also flags:

- Add(*listcomp over e.args);
- an aliased Poly import;
- Poly.___call___;

- as_coefficients_dict;
- a functools.reduce rebuild;
- Symbol(name=...);
- an explicit-constructor rebuild.

A standalone reject of the p19 or p20 gate no longer leaves a stale RELEASE_STATUS.json.

Status. independent_rebuild_passed_p19 is true (bound to the p19 tree and zip; same box). The p20 independent rebuild is pending.

5 P.5 identity step (p19): the projected one-loop κ^2 ch-identity defect

Scope.
p19 keeps the p14r contract and the p15–p18 calculation layers byte-identical (`release_p19.py` recomputes their sha256). It assembles the ch projection (P.3, frozen) of the $\hbar\kappa^2$ part of (S_Ψ, Γ_Ψ) from every evaluated input, exactly in d. It then solves for a symmetry-restoring counterterm in the declared domain. Only the ch projection is covered (one c, one h; one loop; κ^2; off shell). bulk_symmetry_restored is false. O.1 stays NOT_CLOSED; the insertion stays frozen; XV-B and XIV-J are unopened; Route B is not discharged; no $c\,\delta$ is added to u^R_μ.

5.1 Assembly

The ledger `P13_ch_dependency_rows.json` has three surviving rows:

- R_{hh}, $(S_0, \Gamma_1^{[2]})$ $s_2K{:}K.c$, inputs D01+D02+D03+D11+D12, with $X_{hh} = \int (s_0 h)\, G_{hh}\, h$;
- R_{Kc}, $(S_0, \Gamma_1^{[2]})$ $s_1h{:}h^2$, input D13, with $X_{Kc} = \int G_{Kc}\, c\, E\, h$, where E is the h^2-monomial operator of S_0;
- R_{1pt}, $(S_1, \Gamma_1^{[1]})$, inputs D14a–c.

Every input is evaluated; none is Missing, and a Missing input is refused rather than set to zero (control). The defect is computed for seven external assignments, including "c hat, h tilde". For each it is given as the exact d-dependent value times MB, its pole, its coefficient of L and its MS-bar local part.

Result: the defect is zero, exactly in d, in every part and every assignment. Why it is zero. Each surviving row vanishes separately in the projected slot. No counterterm cancellation is involved, and no cancellation between rows.

(i) hh row: full-row transverse decoupling. D01 and D03 are each nonzero in the slot and cancel each other (the p17/p18 transverse sum). D11 is transverse by itself, and D02 = D12 = 0.

(ii) K.c row: D13 reaches the ch slot only through T_1 and E_1 (T_2, T_3 and E_2 are longitudinal; E is transverse), and $A_{T_1} = A_{E_1} = 0$ exactly in d.

(iii) One-point row: the tadpoles D14a–c vanish under the declared prescription.

This rests on:

- M2 / the D11 matching, verified on the *4D components only* (the D11 matching for the evanescent components is not applicable and not verified);

- the evanescent decoupling proofs (P16 E2; P17/P18 hh evanescent and full-row decoupling);
- the frozen P.3 projection and the P13 ledger;
- the declared prescription.

Evanescent decoupling in the assembly. The evanescent parts of both rows were assembled explicitly and are zero for all seven assignments. The evanescent poles themselves are nonzero; they are regulated and not subtracted.

5.2 Counterterm domain and solve

Route G applies $\sigma_0 = (S_0, \cdot)$ to the 11 p14r domain densities: h.h#0–4, c.cb#0–1, c.K#0–2, $\phi.\phi$. Route C computes the covariant ch images exactly in d. Both give rank $P_{ch}\sigma_0 = 3$.

- Over $\mathbb{Q}(i)$ (these counts are the real dimensions under a compatible real structure; wording corrected in p20): rank $\sigma_0|_D = 5$, $\dim D \cap \ker \sigma_0 = 6 = \dim D \cap \operatorname{Im} \sigma_0$, cohomology 0, and the quotient has dimension 5 (complement h.h#0–2, c.cb#0–1).
- The h.h + c.K Γ_1 data, without the ghost self-energy and in route-G (c,cb) labels, have quotient coordinates pole $(0, 0, 0, -3/8, 1/4)$, coefficient of L $(0, 0, 0, 3/8, -1/4)$ and local part $(0, 0, 0, -2/3, 5/12)$. The h.h pole is transverse, i.e. σ_0-exact. These coordinates equal $-[\Sigma]$: Σ itself has $(+3/8, -1/4)$, and the data including Σ have quotient 0 [clarified in p20].
- The linear solve $\Delta + P_{ch}\sigma_0(x) = 0$ gives $x = 0$ (outcome ZERO); no residual and no obstruction.
- The MS-bar counterterm has zero ch image. The h.h + c.K part alone has a nonzero full σ_0 image in the c.b sector; adding the c.cb ghost self-energy counterterm in the correct orientation closes it [corrected in p20, see the p20 erratum note].

Flags and their definitions.

- `projected_defect_computed`: Δ is assembled from all evaluated inputs, exactly in d, in all parts and all seven assignments.
- `projected_identity_verified`: $\Delta = 0$ for the *regulated* Γ_1.
- `projected_restoration_verified` [REDEFINED in p20]: full σ_0 closure, in every ghost +1 sector (h.c, c.b, c.cbs), of the *complete* one-loop κ^2 MS-bar counterterm S_{ct} (h.h + c.K + c.cb, ghost self-energy held fixed, derived relative factor +1, orientation converted). The p19 definition ($P_{ch}\sigma_0(S_{ct}) = 0$, $x = 0$) follows from the identity and locality; it is kept only as a labelled route-consistency control.

Scope of the restoration flag: the one-loop κ^2 counterterm only; not bulk symmetry, not O.1. The release gate re-checks each definition from the certificate content. bulk_symmetry_restored is false regardless.

K1 [corrected in p20; the p19 text is retracted]. p19 reported that full closure of the MS-bar counterterm fails with the derived ghost-equation factor +1 and needs −1. That was a Grassmann-orientation bookkeeping error: the route-G densities are built in the order (c,cb), whereas the P15 Σ coordinates are (cb,c)-oriented, and $V[\text{c.cb}\#k] = -V[(\text{cb}, \text{c})\#k]$. With the measured ratio −1 applied, the derived factor +1 gives full closure (p20 erratum note). The relations below are stated in the (cb,c) orientation:

- The ghost-equation operator $2iD$, verified by P15 as $\Sigma = 2iD[G_{Kc}]$, gives $2iD\,V[\text{c.K}\#1] = -V[(\text{cb}, \text{c})\#0]$ and $2iD\,V[\text{c.K}\#2] = -V[(\text{cb}, \text{c})\#1]$.
- Applied to the c.K pole, this reproduces the P15 Σ pole $(-3/8, 1/4)$ in (cb,c) coordinates, i.e. $(+3/8, -1/4)$ in the route-G (c,cb) coordinates.

J1: finding about the p14r contract (contract not modified). The p14r counterterm-domain certificate counts Im σ_0 3, quotient 8, $D \cap \ker$ 5 and rank $P_{ch}\sigma_0$ 4. These are artefacts of a real/imaginary split over $\mathbb{Q}$: the densities carry an explicit i for odd derivative counts, so an i-multiple counts as independent. Over $\mathbb{Q}(i)$ the counts are 6, 5, 6 and 3 (the real dimensions under a compatible real structure), and Route C agrees. With the split, p19 reproduces the p14r counts (control).

Controls and mutations. All controls are true. Eight mutations each change the outcome:

- drop D03; drop D01; flip the D03 phase; double D03; trace operator instead of $s_0 h$; de Donder $q^2 P$ instead of E; drop the D03 evanescent part (naive 4D). Each gives OBSTRUCTION (nonlocal residual).
- D13 with G_t dropped gives NONZERO but CANCELLABLE (rank 3).

Recorded blind spots of the projected identity: dropping D11 (transverse by itself), dropping D13 ($A_{T_1} = A_{E_1} = 0$) and flipping the K.c row sign (each row vanishes separately). In p20, the full-closure test detects the last two, and the D11 normalisation pin covers the first. Removing D11 from the inventory makes the inventory incomplete, and projected_defect_computed is then not earned.

assignment (c, h)	defect pole	coeff. of L	local finite
`Q4S (declared, R2)`	0	0	0
`QdS (hat)`	0	0	0
`c tilde, h hat`	0	0	0
`c tilde, h mixed`	0	0	0
`c hat, h mixed`	0	0	0
`all tilde`	0	0	0
`c hat, h tilde`	0	0	0

Route G over `Q(i)`: rank $\sigma_0|_D = 5$, dim $D \cap \ker \sigma_0 = 6$, dim $D \cap \mathrm{Im}\, \sigma_0 = 6$, cohomology 0, quotient 5 (complement `h.h#0, h.h#1, h.h#2, c.cb#0, c.cb#1`); rank $P_{ch}\sigma_0 = 3$ (Route C: 3). Over Q with the re/im split (p14r convention): dim $D \cap$ Im = 3, quotient 8, dim $D \cap$ ker = 5.
Quotient coordinates of the Γ_1 local data: pole `['0', '0', '0', '-3/8', '1/4']`, coeff. of L `['0', '0', '0', '3/8', '-1/4']`, local `['0', '0', '0', '-2/3', '5/12']`.

mutation	outcome	changes
`drop_graph_D03`	`OBSTRUCTION (nonlocal residual: the L-coeffi`	yes
`drop_graph_D01`	`OBSTRUCTION (nonlocal residual: the L-coeffi`	yes
`flip_phase_D03`	`OBSTRUCTION (nonlocal residual: the L-coeffi`	yes
`symmetry_factor_D03_x2`	`OBSTRUCTION (nonlocal residual: the L-coeffi`	yes
`wrong_projection_trace_operator_hh`	`OBSTRUCTION (nonlocal residual: the L-coeffi`	yes
`wrong_projection_gauge_fixed_operator_KO`	`OBSTRUCTION (nonlocal residual: the L-coeffi`	yes
`naive_4D_D03_evanescent_dropped`	`OBSTRUCTION (nonlocal residual: the L-coeffi`	yes
`naive_4D_D13_Gt_dropped`	`NONZERO, CANCELLABLE by a counterterm in the`	yes

Counterterm solve: `ZERO; x = 0 (no counterterm needed beyond the MS-bar poles); the identity-preserving finite renormalisations are ker(P_ch sigma_0) cap D (dim 8), of which 2 modulo D cap Im sigma_0.`
Why zero. `each surviving ch row vanishes separately in the projected slot; no counterterm cancellation and no cancellation between rows is involved: (i) hh row (D01+D02+D03+D11+D12): full-row transverse decoupling - D01 and D03 are individually nonzero and cancel against each other in the slot (p17/p18 transverse sum), D11 is transverse by itself, D02 = D12 = 0; (ii) K.c row (D13): D13 reaches the ch slot only through T1 and E1 (T2, T3, E2 are longitudinal, E transverse) and A_T1 = A_E1 = 0 exactly in d; (iii) one-point row: the tadpoles D14a-c vanish under the declared prescription. The defect is therefore zero without any counterterm (solve x = 0).`
Flag definitions. `projected_defect_computed: Delta = P_ch[(S_Psi, Gamma_Psi)]_{hbar kappa^2} assembled from every evaluated input of the three surviving rows (none Missing), exact in d, pole + nonlocal + MS-bar local parts, all seven external assignments.`
`projected_identity_verified: Delta = 0 for the REGULATED one-loop functional Gamma1: pole, nonlocal and local parts vanish for all seven assignments, exactly in d, with all controls true and every mutation changing the outcome.` `projected_restoration_verified: p20 definition (replaces p19's, which follows from the identity + locality): FULL sigma_0 closure of the COMPLETE one-loop kappa^2 MS-bar counterterm S_ct = -(pole) in`

the declared domain, h.h + c.K + c.cb (ghost self-energy Sigma held fixed at its P15 value, orientation converted, derived relative factor +1), in every ghost +1 sector (h.c, c.b, c.cbs), together with projected_identity_verified. Scope: the one-loop kappa^2 counterterm only; NOT bulk symmetry, NOT O.1. **bulk_symmetry_restored**: hard stop: False regardless (not claimed; O.1 NOT_CLOSED).

K1 (corrected in p20; erratum to p19). derived relative factor 1; orientation ratio (route-G (c,cb) over P15 (cb,c)) ['-1', '-1']; Sigma pole in (c,cb) coordinates ['3/8', '-1/4']; full sigma_0 image of the complete MS-bar counterterm: {'closed': True, 'entries': 0, 'sectors': []}. ERRATUM to p19: the p19 statement 'closure fails with the derived factor, needs -1, unresolved' was a Grassmann-orientation bookkeeping error (P15 (cb,c) coordinates placed in route-G (c,cb) slots). Corrected: derived +1, full closure holds (finding K1 of the p19 independent rebuild).

J1 (finding about the p14r contract). FINDING ABOUT THE p14r CONTRACT (contract NOT modified): the p14r counterterm-domain counts (Im sigma_0 3, quotient 8, D cap ker 5, P_ch sigma_0 rank 4) are artefacts of a re/im split over Q (the densities carry explicit i for odd derivative counts, so i-multiples count as independent). The counts over Q(i) - sigma_0 rank 5, D cap ker 6, D cap Im 6, cohomology 0, quotient 5, P_ch sigma_0 rank 3 (= Route C) - equal the real dimensions under a compatible real structure. The p14r certificate stays as released; both are reported

6 P.5 calculation, step 4 (p18): M2 discharged, D11 evaluated, the full hh slot

Scope.
p18 keeps the p14r contract and the p15, p16 and p17 calculation layers byte-identical (`release_p18.py` recomputes their sha256). It discharges the D11 matching premise M2 on the 4D components, evaluates D11 under the declared prescription, and rechecks the ch-projected hh slot with D11 included. quantum_inputs_evaluated is earned (every graph of the inventory is evaluated in one prescription). projected_identity_verified, projected_restoration_verified and bulk_symmetry_restored are NOT earned. O.1 stays NOT_CLOSED; the insertion stays frozen; XV-B and XIV-J are unopened; Route B is not discharged; no $c\,\delta$ is added to u^R_μ.

6.1 M2 (certificate `P18_D11_M2.json`)

M2 states that on $U = \mathbb{R}^4 \setminus \{0\}$ the inverse Fourier transform of P^{DR}_{AB} equals $R_A R_B\,\Pi^R_{\rm tr}$, with $\Pi^R_{\rm tr} = (i\kappa^2/4)\,K u^R_\mu$, $K = \alpha D_2 + \beta D_0$, $\alpha = A^2/60$, $\beta = A^2/18$ and $A = 1/4\pi^2$.

- *Route B (exact; primary).* Write $P^{DR} = \kappa^2 N_\varepsilon \hat{G}$ with $\hat{G} = \sum_s A_s(d) S_s\,\mathrm{MB}$ and $\mathrm{MB} = 1/\varepsilon + 2 - L$. At $d \to 4$ its nonlocal part is $-(i\kappa^2/16\pi^2) L \sum_s A_s(4) S_s(p)$; everything else is polynomial. On the other side, $\mathcal{F}[u^R] = c_{\rm FT}\log(-p^2 - i0) + \mathrm{const}$, and $K \to \hat{K}(p)$ with $\hat{P}_{mn} = p_m p_n - \eta_{mn} p^2$. This operator is fourth order, so the sign convention of the transform drops out. M2 then holds iff, for each of the 100 component pairs, $X_{AB} = \sum_s A_s(4) S_s|_{AB} = -4\pi^2 c_{\rm FT} A^2\, Y_{AB} = -(i/4)\, Y_{AB}$, where $Y = R_A R_B(\hat{D}_2/60 + \hat{D}_0/18)$. This holds exactly on all 100 pairs. The difference then has a polynomial Fourier transform, so it is supported at $\{0\}$, and μ enters only through polynomials.

- $c_{\rm FT}$*, calibrated twice.* (a) From the library DR bubble, $\mathcal{F}[D_F^2] = -N_\varepsilon \mathrm{MB}$ and $D_F^2 = A^2 u$. (b) As $-r$, with the XVI-O residue $r = -i\pi^2$. Both give $c_{\rm FT} = i\pi^2$.

- *Route A (structural corroboration).* M1 is re-run: the P integrand equals the Fourier convolution of the XVI-O product $T = VV(CC + CC)$ (100/100). The frozen XVI-O code, imported read-only, re-derives α, β and verifies $T = Ku$ on U for all 256 components. A regularised convolution agrees with the pointwise product on U (cited: XVI-O lem:BV_rules_on_U, Constructed).

- *Controls and mutations.* D11 is transverse exactly in d; the FP and PV loop routes agree. Six mutations are each rejected: $\alpha \leftrightarrow \beta$, off-diagonal weight 1, flipped FT sign, doubled symmetry factor, D01 in place of D11, and $\beta = 0$.

- *Scope.* The flag is qualified as D11_matching_verified_scope = "4D components". M2 is shown on the 4D components only. The evanescent components are the 8 $\tilde{g}$-carrying structures, i.e. components with an index pair in the ε-scalar directions. Their status is not_applicable_not_verified, because XVI-O is strictly 4D and has no counterpart. The evanescent structures of D11 have nonzero L-coefficients (8, recorded); they are ε-scalar components outside the 4D statement.

- *Consequence (M3/M4).* M3 is now unconditional on the 4D components: L_{match} is supported at 0. It is transverse and homogeneous of degree four, so it lies in the 2-dimensional transverse h·h subspace. Its coordinates need the XVI-O finite constant of u^R and are not computed.

6.2 D11 (certificate `P18_D11.json`) and the full slot (`P18_hh_slot.json`)

- *Prescription.* Operational DRED R1–R6 with opaque d, MS-bar, Feynman $+i0$, N4 contact exclusions, M3 bounds 4/8/4, exactly-degree-four normalisation. Both points are checked per structure: every structure has momentum degree 4, so the order-0/2 contacts excluded by N4 do not arise.

- *Inventory.* quantum_inputs_evaluated is earned. Every graph of the three surviving ch rows is evaluated, and none is Missing: hh D01, D02, D03, D11, D12; K·c D13; one-point D14a–c.

- *Evaluation.* 14 structures, 6 of them 4D. The covariant form is exact and unique; the FP and PV routes agree exactly in d; the result is exchange symmetric. The nonlocal coefficient equals minus the pole. Premature $d = 4$ is refused and would shift the local parts. Each of the four mutations is rejected.

- *4D pole.* $z_{D11} = i(-1/160, -1/480, 1/80, 1/240, -1/120) = i(\frac{1}{240}k_1 - \frac{1}{120}k_2)$, which is transverse.

- *Full row.* $p_m G_{D11}^{(mn)(rs)} = 0$ holds identically in d, for the 4D and the evanescent parts. D11 therefore decouples from the slot by itself; D01 and D03 decouple only as a sum.

  - $X_{D01+D03+D11} = 0$ for all seven external assignments. This includes $\hat{c}, \tilde{h}$ (H2), where $X_{D01} = -X_{D03} = p^4\, \tilde{a}{\cdot}\tilde{b}\; \hat{c}{\cdot}p/16$.
  - No evanescent structure survives, so D11 does not reopen the evanescent question and no counterterm revision is needed.
  - The summed 4D pole is $i(\frac{41}{80}k_1 - \frac{43}{120}k_2)$, which is transverse.
  - Caveat: D11 phase and symmetry-factor mutations are invisible to the slot. They are tested by M2 instead.
  - The summed evanescent poles of the hh Green function (8 structures) are nonzero but transverse. Renormalising that Green function itself is outside the projection and not claimed.

M2: `DISCHARGED (4D components)`; route B: 100 component pairs, 0 mismatches, constant `-I/4`; c_FT = `I*pi^2` (DR bubble) = `I*pi^2` (-r).

D11 structure	evan.	UV pole	coeff. of L	local (MS-bar)	shift (d=4 first)
g4_mn g4_rs (p.p)^2	no	-I/80	I/80	-I/75	7*I/600
g4_mn gt_rs (p.p)^2	yes	-I/120	I/120	-77*I/3600	-17*I/3600
g4_mn p_r p_s (p.p)^1	no	I/80	-I/80	I/75	-7*I/600
g4_mr g4_ns (p.p)^2	no	-I/480	I/480	-23*I/3600	-I/450
g4_mr gt_ns (p.p)^2	yes	I/480	-I/480	23*I/3600	I/450
g4_mr p_n p_s (p.p)^1	no	I/480	-I/480	23*I/3600	I/450
g4_rs gt_mn (p.p)^2	yes	-I/120	I/120	-77*I/3600	-17*I/3600
g4_rs p_m p_n (p.p)^1	no	I/80	-I/80	I/75	-7*I/600
gt_mn gt_rs (p.p)^2	yes	-I/480	I/480	-23*I/3600	-I/450
gt_mn p_r p_s (p.p)^1	yes	I/120	-I/120	77*I/3600	17*I/3600
gt_mr gt_ns (p.p)^2	yes	-I/480	I/480	-23*I/3600	-I/450
gt_mr p_n p_s (p.p)^1	yes	-I/480	I/480	-23*I/3600	-I/450
gt_rs p_m p_n (p.p)^1	yes	I/120	-I/120	77*I/3600	17*I/3600
p_m p_n p_r p_s	no	-I/60	I/60	-47*I/1800	13*I/1800

4D pole	z_1	z_2	z_3	z_4	z_5	transverse
D11	-I/160	-I/480	I/80	I/240	-I/120	yes
D01+D03	-23*I/240	-61*I/240	23*I/120	61*I/120	-7*I/20	yes
D01+D03+D11	-49*I/480	-41*I/160	49*I/240	41*I/80	-43*I/120	yes

assignment (c, h)	X_{D01}	X_{D03}	X_{D11}	sum
Q4S (declared, R2)	p.p^2*(d*a.b*c^.p - 3*a.b*c^.p + a.c*b^.p + a^.p*b.c - a^.p*b .c + 2*a .b *c^.p - a .c *b^.p)/(16*(d - 1))	-p.p^2*(d*a.b*c^.p - 3*a.b*c^.p + a.c*b^.p + a^.p*b.c - a^.p*b .c + 2*a .b *c^.p - a .c *b^.p)/(16*(d - 1))	0	0
QdS (hat)	p.p^2*(d*a^.b^*c^.p - 3*a^.b^*c^.p + a^.c^*b^.p + a^.p*b^.c^)/(16*(d - 1))	-p.p^2*(d*a^.b^*c^.p - 3*a^.b^*c^.p + a^.c^*b^.p + a^.p*b^.c^)/(16*(d - 1))	0	0
c tilde, h hat	0	0	0	0
c tilde, h mixed	0	0	0	0
c hat, h mixed	0	0	0	0
all tilde	0	0	0	0
c hat, h tilde	a .b *c^.p*p.p^2/16	-a .b *c^.p*p.p^2/16	0	0

Full-row verdict: `DECOUPLED (proof, full row): p_m G = 0 exactly in d for D11 by itself and for D01 + D03 as a sum; no evanescent structure survives in the slot.`

7 Scope of revision p17: the hh evanescent poles and the ch slot

p17 = P.5 calculation, third step; D11 is NOT evaluated; no slot is complete.

p17 keeps the p14r contract and the p15 and p16 calculation layers (byte-identical) and adds: (1) a deterministic analysis of whether the evanescent ($\tilde{g}$) UV poles of D01 and D03 reach the ch-projected slot, with controls and mutations; (2) the explicit coordinates of the 4D poles in the p14r $h \cdot h$ basis and the constraint they satisfy; (3) the fixes G1–G4 and G6 of the p16 independent review (one disclosed dred.py revision: the guarded loader `parse_d`). D11 stays NOT evaluated (M2 undischarged), `D11_matching_verified`=false and `quantum_inputs_evaluated` stays not earned. O.1 is NOT_CLOSED, `symmetry_restored`=`bulk_symmetry_restored`=false, the insertion stays frozen, XV-B and XIV-J stay closed, Route B is not discharged, XVI-P is unregistered, and no $c\delta$ term is added to u^R_μ.

7.1 Where the hh self-energy enters the ch projection

Reading the ledger `ledgers/P13_ch_dependency_rows.json` (a computed control, not an assertion), the graphs D01, D03 occur in exactly one row: $(S_0, \Gamma_1^{[2]})\, s_2\, K{:}K{\cdot}\, c$, with Γ_1 pattern h^2. The same row also holds D02, D12 (zero, p15) and D11 (not evaluated). The slot functional is

$$X_{hh}[c, h] = \int (s\, h)_{(mn)}(p; c)\; G^{(mn)(rs)}(p)\; h_{(rs)}(-p).$$

The operator $s\, h$ is read off the $K \cdot c$ monomial of the frozen $S_\Psi^{(0)}$ (16 nonzero components). Its covariant fit is exact and unique: $(s\, h)_{(mn)\nu} = \frac{i}{2}(\eta_{m\nu} p_n + \eta_{n\nu} p_m)$. It is therefore LONGITUDINAL in (mn).

7.2 Decoupling (proof, exact in d)

- **H1.** $s\, h$ is longitudinal (above), so X_{hh} depends on G only through $p_m G^{(mn)(rs)}$. Because $p \in$ QdS, $p \cdot \tilde{g} = 0$, so $\tilde{g}$ on the pair (mn) drops out.
- **H2/H3.** $p_m\, (G_{D01} + G_{D03})^{(mn)(rs)} = 0$ identically in d. This holds for the 4D part and for the evanescent part separately: each of the six structures of $p_m G$ (for example $\tilde{g}_{rs} p_n$: $-\frac{i\, p^4}{8(d-1)}$ from D01 and $+\frac{i\, p^4}{8(d-1)}$ from D03) cancels exactly. Hence $X_{D01+D03} = 0$ for all six external assignments (Q4S, QdS, mixed $\hat{}/$). With $D02 = D12 = 0$, the hh slot receives nothing from the evaluated graphs.

- **H4 (controls).** X_{D01} and X_{D03} are each nonzero, and so are their evanescent parts alone. The cancellation is therefore a property of the SUM (the ghost loop cancels the longitudinal part of the bubble), not of each graph. The map is not trivially zero.
- **Order.** At one loop the ch projection is linear in Γ_1, so no product of evanescent poles occurs. The cancellation is exact in d and not merely $O(\epsilon)$.

Mutations, each rejected (summed slot becomes nonzero): wrong relative sign of D03; D03 doubled (symmetry factor); D03 with its evanescent part dropped (naive 4D); the trace operator $\eta_{mn}p_\nu$ in place of $s\,h$.

Verdict: DECOUPLED (proof), as the sum D01 + D03 (+ D02 = D12 = 0), and not graph by graph. No evanescent structure survives in the slot. No evanescent $h \cdot h$ counterterm is added, so the counterterm quotient is unchanged and no contract file changes for this. The summed evanescent poles of the hh Green function are nonzero (eight structures, recorded in the certificate) but transverse. Renormalizing the hh Green function itself would need evanescent $h \cdot h$ counterterms; that matters only outside this projection and is not claimed. D11, in the same row, is open.

7.3 4D poles: coordinates and the transversality constraint

The five p14r $h \cdot h$ directions span the five pair-symmetric 4-derivative 4D structures (rank 5), so span membership is automatic. The sharper statement is a constraint. Longitudinality $p_m G^{(mn)(rs)} = 0$ imposes rank-3 linear conditions on the coordinates z, which leaves a 2-dimensional transverse subspace with basis $k_1 = (\frac{1}{2}, -\frac{1}{2}, -1, 1, 0)$ and $k_2 = (1, 0, -2, 0, 1)$. The individual pole coordinates are not transverse:

$$z_{D01} = i\,(-\tfrac{27}{160}, -\tfrac{71}{240}, \tfrac{19}{60}, \tfrac{11}{20}, -\tfrac{13}{30}), \qquad z_{D03} = i\,(\tfrac{7}{96}, \tfrac{1}{24}, -\tfrac{1}{8}, -\tfrac{1}{24}, \tfrac{1}{12}).$$

Their sum is transverse:

$$z_{D01+D03} = i\,(-\tfrac{23}{240}, -\tfrac{61}{240}, \tfrac{23}{120}, \tfrac{61}{120}, -\tfrac{7}{20}) = i\,(\tfrac{61}{120}\,k_1 - \tfrac{7}{20}\,k_2).$$

This is the Ward-type statement the ch identity needs at this order. It is incomplete without D11.

7.4 Phase convention (G6)

Every graph is exported as $G = \kappa^2 N_\epsilon \hat{G}$ with $N_\epsilon = iS_\epsilon/16\pi^2$. For D01 and D03, $\hat{G} = -i\,\mathrm{pref}/\kappa^2 \int$ with real export prefactors $(\kappa^2/2, -\kappa^2)$, so $A(d)$ carries an overall i. D13's exported prefactor is $i\kappa^2$, so its $A(d)$ is real. Because $s\,h = i(\ldots)$, X_{hh} is real like the D13 slot. The D01/D03 relative phase is tested by H3 (mutation M1 breaks it). The D13-versus-hh relative phase is not tested by a nonzero identity, since both slot contributions vanish. No physical overall phase is asserted.

7.5 Review fixes G1–G4

G2: `dred.parse_d` (locals $\{d$: opaque $d\}$) and `dred.assert_opaque_d` close the plain-`Symbol('d')` hazard of re-parsed strings (disclosed contract revision; no value affected). G1/G3: the guard docstring and the disclosure now state exactly what is refused at runtime and what is excluded by the extended static AST scan: replace with any function query, sympify/S/parse_expr of non-sympy data without the guarded locals, builtin eval/exec, srepr round-trips, `.func(*args)` rebuilds, `Poly(..).eval`, lambdify. The scan is a pattern check, not a proof. G4: the change log lists the p17 reference-values file as added.

structure of $G^{(mn)(rs)}$	evanescent	D01 reaches slot	D03 reaches slot
`g4_mn g4_rs (p.p)^2`	no	yes	yes
`g4_mn gt_rs (p.p)^2`	yes	yes	yes
`g4_mn p_r p_s (p.p)^1`	no	yes	yes
`g4_mr g4_ns (p.p)^2`	no	yes	yes
`g4_mr gt_ns (p.p)^2`	yes	yes	yes
`g4_mr p_n p_s (p.p)^1`	no	yes	yes
`g4_rs gt_mn (p.p)^2`	yes	no	no
`g4_rs p_m p_n (p.p)^1`	no	yes	yes
`gt_mn gt_rs (p.p)^2`	yes	no	–
`gt_mn p_r p_s (p.p)^1`	yes	no	no
`gt_mr gt_ns (p.p)^2`	yes	no	–
`gt_mr p_n p_s (p.p)^1`	yes	yes	yes
`gt_rs p_m p_n (p.p)^1`	yes	yes	yes
`p_m p_n p_r p_s`	no	yes	yes

4D pole	z_1	z_2	z_3	z_4	z_5	transverse
D01	`-27*I/160`	`-71*I/240`	`19*I/60`	`11*I/20`	`-13*I/30`	no
D03	`7*I/96`	`I/24`	`-I/8`	`-I/24`	`I/12`	no
D01+D03	`-23*I/240`	`-61*I/240`	`23*I/120`	`61*I/120`	`-7*I/20`	yes

Verdict: `DECOUPLED (proof): as the sum D01 + D03 (+ D02 = D12 = 0); not graph by graph`. Surviving evanescent structures in the slot: none. Transverse kernel (p14r coordinates): `(1/2, -1/2, -1, 1, 0); (1, 0, -2, 0, 1)`. D11: NOT evaluated.

8 Scope of revision p16: E2 analysis and Batch C

p16 = P.5 calculation, second step; D11 is NOT evaluated; no slot is complete.

p16 keeps the p14r contract and the p15 calculation layer (byte-identical) and adds: (1) a deterministic proof that the evanescent D13 pole (structure E2) decouples from the ch-projected slot, with mutation controls; (2) Batch C, the graphs D01 and D03 of the $\kappa^2\, hh$ slot, under the same frozen prescription (DRED R1–R6, Feynman $+i0$, MS-bar, S0, d opaque); (3) the fixes F3–F5 of the p15 independent review. D11 stays NOT evaluated (M2 undischarged), `D11_matching_verified`=false and `quantum_inputs_evaluated` stays not earned. O.1 is NOT_CLOSED, `symmetry_restored`=`bulk_symmetry_restored`=false, the insertion stays frozen, XV-B and XIV-J stay closed, Route B is not discharged, XVI-P is unregistered, and no $c\delta$ term is added to u^R_μ.

8.1 E2: decoupling from the ch-projected slot (proof)

In p15 D13 has an evanescent structure $E2_{(mn)\nu} = (\tilde g_{m\nu} p_n + \tilde g_{n\nu} p_m) p^2$ with $A_{E2}(d) = 3/(16(d-1))$ and UV pole 1/16; p14r declares no evanescent counterterm. D13 enters the ch projection only through the row $(S_0, \Gamma_1^{[2]})\, s_1\, h{:}h^2$:

$$X_{D13}[c,h] = \int G_{(mn)\nu}(p)\, c^\nu(p)\, E^{(mn)(rs)}(p)\, h_{(rs)}(-p),$$

where E is the operator of the h^2 monomials of $S^{(0)}_\Psi$ (linearized Einstein–Hilbert). Its exact covariant form is fitted from the frozen density and verified on all components; the fit is unique.

- **L1.** $p_m E^{(mn)(rs)} = 0$ as a covariant identity. Only $g_4 \cdot p = p$ is used; no trace is taken, so the identity holds exactly in d.
- **L2.** $E2$ is longitudinal in (mn).
- **L3.** Therefore $E2 \cdot E = 0$ identically. The slot value is 0 for every external assignment (Q4S, QdS, mixed $\hat{}\tilde{}$ components), exactly in d and not merely $O(\epsilon)$.
- **L4 (control).** The non-longitudinal $T1$ and $E1$ give nonzero slot values, so the map is not trivially zero. $A_{T1} = A_{E1} = 0$ for D13, so D13's whole ch-slot contribution is 0.

Mutation controls: the gauge-fixed operator $q^2 P$ in place of E, $E1$ in place of $E2$, and a wrong index slot each make the $E2$ slot nonzero and are rejected. At one loop the ch projection is linear in Γ_1, so no product of evanescent poles arises.

Outcome: DECOUPLED (proof). No evanescent counterterm is added and no contract file changes for E2. The $E2$ pole remains a UV pole of the ϵ-scalar component of the $K' \cdot c$ Green function (R6: retained and recorded). Its renormalization matters only outside this projection, and it is not claimed here. D13 stays *regulated*, with a ch-slot contribution of 0.

8.2 Batch C: D01, D03 (covariant route)

Finding C1. For rank-4 numerators, the p14r per-component R3 lift produces component sums quadratic in $\tilde{g}$ (D03: 10 of 100 components). After the component sums, their abstract value cannot be recovered, because the linear realization $\tilde{g} = (4-d)M$ is not faithful there. A finite component basis cannot carry a projector of non-integer trace.

Batch C therefore uses a *covariant route* that applies the same rules R1–R6 with free external indices only:

1. The exact covariant 4D numerator is fitted from the frozen export and verified as a polynomial identity on all 100 components. It is unique (upper external indices, off-diagonal weight 2).
2. The lift $\eta \to g_4$ is made.
3. Two independent loop routes are used: Feynman parameters with symmetric integration, and the frozen `dred.reduce_tensor` with the J library and S0. They agree exactly in d.
4. Then $\hat{g} = g_4 - \tilde{g}$ on the free indices.

D13 is reproduced exactly by this route (control), and the rules are unchanged (no contract change).

Further controls:

- The integrands are regenerated from S_Ψ (D01 from the ℓ_3 density, D03 from the ghost vertices) and equal the export on 100 of 100 components.
- The results are exchange symmetric.
- The nonlocal log coefficient equals minus the pole.
- The 4D poles lie in the span of the five p14r $h.h$ counterterm directions.

Seven mutations are each rejected: ghost-loop sign, symmetry factor, premature $d = 4$, dropped $\tilde{g}$, external index position, off-diagonal weight, and the component route with the linear realization.

graph	structure	$A(d)$ (coefficient of MB)	UV pole	MS-bar finite
D01	g4_mn g4_rs (p.p)^2	I*(-2*d^2 - 10*d - 9)/(16*(d^2 - 1))	-27*I/80	I*(405*L - 982)/1200
D01	g4_mn gt_rs (p.p)^2	I*(5*d + 6)/(16*(d^2 - 1))	13*I/120	I*(523 - 195*L)/1800
D01	g4_mn p_r p_s (p.p)^1	I*(d^2 + 12*d + 12)/(16*(d^2 - 1))	19*I/60	I*(724 - 285*L)/900
D01	g4_mr g4_ns (p.p)^2	I*(-6*d^2 + 4*d + 9)/(16*(d^2 - 1))	-71*I/240	I*(1065*L - 1946)/3600
D01	g4_mr gt_ns (p.p)^2	I*(-2*d - 1)/(16*(d^2 - 1))	-3*I/80	I*(45*L - 118)/1200
D01	g4_mr p_n p_s (p.p)^1	I*(6*d^2 - 5*d - 10)/(16*(d^2 - 1))	11*I/40	I*(97 - 55*L)/200
D01	g4_rs gt_mn (p.p)^2	I*(5*d + 6)/(16*(d^2 - 1))	13*I/120	I*(523 - 195*L)/1800
D01	g4_rs p_m p_n (p.p)^1	I*(d^2 + 12*d + 12)/(16*(d^2 - 1))	19*I/60	I*(724 - 285*L)/900
D01	gt_mn gt_rs (p.p)^2	-I/(16*d^2 - 16)	-I/240	I*(15*L - 46)/3600
D01	gt_mn p_r p_s (p.p)^1	I*(-7*d - 8)/(16*(d^2 - 1))	-3*I/20	I*(90*L - 241)/600
D01	gt_mr gt_ns (p.p)^2	-I/(16*d^2 - 16)	-I/240	I*(15*L - 46)/3600
D01	gt_mr p_n p_s (p.p)^1	I*(3*d + 2)/(16*(d^2 - 1))	7*I/120	I*(277 - 105*L)/1800
D01	gt_rs p_m p_n (p.p)^1	I*(-7*d - 8)/(16*(d^2 - 1))	-3*I/20	I*(90*L - 241)/600
D01	p_m p_n p_r p_s	I*(-13*d^2 - 2*d + 8)/(16*(d^2 - 1))	-13*I/15	I*(780*L - 1597)/900
D03	g4_mn g4_rs (p.p)^2	I*(d + 3)/(16*(d - 1))	7*I/48	I*(50 - 21*L)/144
D03	g4_mn gt_rs (p.p)^2	-I/(8*d - 8)	-I/24	I*(3*L - 8)/72
D03	g4_mn p_r p_s (p.p)^1	-3*I/(8*d - 8)	-I/8	I*(3*L - 8)/24
D03	g4_mr g4_ns (p.p)^2	I*(d - 3)/(8*(d - 1))	I/24	I*(2 - 3*L)/72
D03	g4_mr gt_ns (p.p)^2	I/(8*(d - 1))	I/24	I*(8 - 3*L)/72
D03	g4_mr p_n p_s (p.p)^1	I*(7 - 2*d)/(16*(d - 1))	-I/48	I*(3*L + 4)/144
D03	g4_rs gt_mn (p.p)^2	-I/(8*d - 8)	-I/24	I*(3*L - 8)/72
D03	g4_rs p_m p_n (p.p)^1	-3*I/(8*d - 8)	-I/8	I*(3*L - 8)/24
D03	gt_mn p_r p_s (p.p)^1	I/(4*(d - 1))	I/12	I*(8 - 3*L)/36
D03	gt_mr p_n p_s (p.p)^1	-3*I/(16*d - 16)	-I/16	I*(3*L - 8)/48
D03	gt_rs p_m p_n (p.p)^1	I/(4*(d - 1))	I/12	I*(8 - 3*L)/36
D03	p_m p_n p_r p_s	I*(d - 2)/(4*(d - 1))	I/6	I*(5 - 3*L)/18

Status: D01: regulated. D03: regulated. E2 (D13): DECOUPLED (proof).

With $A(d)$ the coefficient of MB $= 1/\epsilon + 2 - L$, $L = \log(-p^2/\mu^2 - i0)$, in units of N_ϵ: the UV pole is $A(4)$, and the MS-bar finite part is $A(4)(2-L) - 2A'(4)$. *Status:* D01 and D03 are *regulated*. Their 4D sectors are renormalized by poles in the p14r $h.h$ span. Their evanescent ($\tilde{g}$) poles are NONZERO and retained, and no evanescent $h.h$ counterterm is declared. The hh slot (D01+D02+D03+D11+D12) stays incomplete because D11 is not evaluated.

8.3 Review fixes F3–F5 and rebuild flags

- **F3 (one disclosed contract revision, `dred.py`).** The opaque d now refuses `expr.replace(d,·)`, pattern matching on d and the lambdify/code-printer paths outside `substitute_d`. Paths without a hook (callable-query replace, `Poly.eval`, string round trips) are excluded by a static AST scan. The P13–P15 certificate values are unchanged against the p15 reference values.
- **F4.** The page footer comes from one version variable. The PDF readback also covers the D13, Batch A, Batch C and E2 value blocks.
- **F5.** Every changed file, contract and non-contract, is listed in `change_log_vs_p15` of the release status.

`independent_rebuild_passed_p15`=true is bound to the p15 tree 79f53752… and zip 54325c1f… (same box). The p16 tree's own independent rebuild is pending.

9 Scope of revision p15: first P.5 calculation

p15 = P.5 calculation, first batch; D01, D03, D11 are NOT evaluated; no slot is complete.

p15 evaluates the Batch A graphs D02, D12, D14a, D14b, D14c and the Batch B graph D13 under the frozen p14r prescription: DRED R1–R6 with the $\hat{g}/\tilde{g}$ algebra, contractions before the Laurent expansion, d kept symbolic and evanescent parts retained; Feynman $+i0$; MS-bar with $N_\epsilon = iS_\epsilon/(16\pi^2)$, $S_\epsilon = (4\pi)^\epsilon e^{-\gamma_E \epsilon}$. The declared action is massless (no ϕ mass is declared). `quantum_inputs_evaluated`, `projected_defect_computed`, `projected_identity_verified` and `projected_restoration_verified` remain not earned, and `bulk_symmetry_restored`=false. O.1 is NOT_CLOSED, the insertion stays frozen, XV-B and XIV-J stay closed, and Route B is not discharged. The new flags `batchA_evaluated` and `D13_evaluated` each carry an execution, a claim and a scope part.

9.1 Integral library and the scaleless rule

The library (`P15_integral_library.json`) provides A_0, B_0 and the tensor reduction in $\hat{g}$. It is certified exactly in d against Feynman parameters, and numerically at $d = 2.6, 3.3, 3.7$.

The scaleless rule is *stated*, not assumed:

$$\text{S0:} \quad \int d^d l \, (l^2)^k/(l^2)^\nu := 0, \qquad \omega = 4 + 2k - 2\nu.$$

Its UV/IR reading is as follows. For $\omega = 0$ the zero is the cancellation $1/\epsilon_{\rm UV} - 1/\epsilon_{\rm IR}$. For $\omega \neq 0$ it discards a pure power divergence, and neither the UV part nor the IR part has a pole at $d = 4$. Applying S0 to an integral that carries a scale is an error, and Missing is never replaced by zero by convention.

9.2 Batch A: D02, D12, D14a–c

All five graphs are *vanishing under the declared prescription*: there is no UV pole, the finite part is 0, and no p14r counterterm is needed. Every scaleless member is a pure power divergence ($\omega = 2, 4$), so the zero is *not* a $1/\epsilon_{\rm UV} - 1/\epsilon_{\rm IR}$ cancellation.

- **IR-regulator mass** (analysis only). With a mass the poles are proportional to $\kappa^2 m^4 hh$ and $\kappa^2 m^2 p^2 hh$, which are order-0/2 contacts excluded by N4, or to $\kappa m^4 h$, for which the κ^1 basis is empty. All of them vanish as $m \to 0$.
- **Analytic regulator.** It does not separate the UV from the IR here.
- **Cross-check.** Two routes agree: the frozen dred lift with S0, and an independent symmetric tensor integration.
- **D12.** D12 ($\kappa^2\, hh$) is kept separate from the order-κ one-point slot (D14a–c). `D12_status` moves from Missing to *vanishing_under_declared_prescription*. This is *derived* under P's own prescription; XVI-O's value is not imported.

9.3 Batch B: D13 (κ^2 $\mathbf{K'}\cdot c$)

Write $G_{(mn)\nu}(p) = V_{\Gamma_1}[K'_{(mn)}(-p), c^\nu(p)] = \kappa^2 N_\epsilon \hat{G}$, with $\hat{G} = \sum_i A_i(d)\, T_i\, B_0$, $B_0 = 1/\epsilon + 2 - L$ and $L = \log(-p^2/\mu^2 - i0)$. The structures are

$$T_1 = \eta_{mn} p_\nu p^2, \quad T_2 = (\eta_{m\nu} p_n + \eta_{n\nu} p_m) p^2, \quad T_3 = p_m p_n p_\nu,$$

together with the evanescent $E_1 = \tilde{G}_{mn} p_\nu p^2$ and $E_2 = (\tilde{G}_{m\nu} p_n + \tilde{G}_{n\nu} p_m) p^2$. The coefficients are

$$A_{T_2} = \frac{4d-7}{16(d-1)}, \qquad A_{T_3} = \frac{3(2-d)}{8(d-1)}, \qquad A_{E_2} = \frac{3}{16(d-1)}, \qquad A_{T_1} = A_{E_1} = 0.$$

This gives:

- **UV poles:** T_2: 3/16, T_3: $-1/4$, E_2: 1/16.
- **MS-bar finite parts:** T_2: $1/3 - \frac{3}{16}L$, T_3: $-5/12 + \frac{1}{4}L$, E_2: $1/6 - \frac{1}{16}L$. The nonlocal part is $-(\text{pole})\cdot L$, and the local part is a cubic polynomial.
- **Counterterm map.** The 4D pole lies in the span of the three p14r $c\cdot K'$ directions, with coordinates $(0, 3/8, -1/4)$. The local ambiguity is a finite renormalization along the same directions.
- **d-dependence.** The $(d-4)\times$pole terms shift the local part (T_2: $-1/24$, T_3: $+1/12$). Setting $d = 4$ first gives a different MS-bar finite part.
- **Evanescent pole and status.** The evanescent pole (E_2) is nonzero, and the p14r contract declares no evanescent counterterm. D13's status is therefore *regulated*. Its 4D sector is *renormalized*; the evanescent pole is recorded but not absorbed, and stays an open contract item.

Controls.

- **Two routes, exact in d.** Route 1 is the sha-pinned export through the dred Pipeline. Route 2 is the integrand regenerated from the rule tables and evaluated by Feynman parameters with d symbolic. They agree exactly.
- **Exact covariant fit.**
- **BRST sanity (ghost equation).** $2i\, D^\rho[G]$ equals the evaluated one-loop ghost self-energy. The ghost self-energy's 4D pole lies in the p14r $c\cdot\bar{c}$ span, at $(-3/8, 1/4)$; its evanescent pole is $-1/8$.
- **Structural R5 safeguard.** It is active.

Mutations. Each mutation changes the result and is rejected: wrong ghost-loop sign, premature $d = 4$ (refused at the substitution), dropped evanescent part, wrong symmetry factor, and wrong K'/c ordering.

graph	status	UV pole (units $\kappa^2 N_\epsilon/\epsilon$)	finite part
D02	vanishing_under_declared_prescription	0	0
D12	vanishing_under_declared_prescription	0	0
D14a	vanishing_under_declared_prescription	0	0
D14b	vanishing_under_declared_prescription	0	0
D14c	vanishing_under_declared_prescription	0	0
D13	regulated (4D: renormalized)	T1: 0, T2: 3/16, T3: -1/4, E1: 0, E2: 1/16	local T1: 0, T2: 1/3, T3: -5/12, E1: 0, E2: 1/6

9.4 Disclosed p14r contract revision

Finding F1 of the p14r independent rebuild concerns the p14r `PrematureD4Substitution` guard in `dred.py`, which was a heuristic.

- **What went wrong.** It gave a false positive on $\eta^{rs}\int l_r l_s/(l^2 - m^2) = J$, a correct d-free result. It gave a false negative under the premature-$d = 4$ mutation on the same integral: the result was $dJ/4$, a finite error of $-m^4/2$.
- **What p15 does.** p15 replaces the guard by a structural, provenance-based safeguard. d is opaque and can be substituted only through `dred.substitute_d`, which records each use. Numeric traces of $\hat{g}/\tilde{g}$ are refused. The R5 docstring is corrected, and the reviewer's toys are regression tests.
- **What does not change.** R1–R6 are unchanged, and every other contract file is byte-identical (checked against pinned sha256). Released p14r values are unaffected, since p14r applied the Laurent step only to toys.

9.5 Reproducibility

The independent rebuild of p14r *passed*: 143/143 values equal and 60 tests. It is bound to the p14r tree `a64c0ae3...` and zip `19e8bf59....` It ran on the same box, in a fresh directory and venv, and was not run on a second machine. It is not inherited by p15, whose own independent flag is pending.

10 Scope of revision p14 (repaired in p14r)

p14r = a narrow repair of p14; P.3/P.4 code byte-identical; no loop evaluated.

- The prescription is made operational: executable DRED index rules, tests against premature $d = 4$, and the withdrawn claim that no evanescent remainder arises (Section 14.1).
- The D11 M3 labels are corrected, and M3 is made conditional on M2, with an explicit normalization condition.
- The counterterm quotient is worded as candidate directions before the identity constraints.
- The regeneration coverage is stated, and the independent Hessian-route regeneration W6 is added for D01, D02, D03 and D12.
- The isolated-copy gate tests contain the frozen inputs.

p14 is archived in `archival/p14/`.

p14 = P.5 entry: the contract is fixed; no loop is evaluated.

p14 fixes what P.5 will compute and how: the quantum object Γ_Ψ in transformed variables (Section 13); one prescription together with a D11 matching note against XVI-O's frozen D11 (Section 14); the one-loop counterterm domain at κ^1, κ^2 (Section 15); and executable amplitude rules with symbolic integrands for D01, D02, D03, D11 (reference only), D12, D13, D14a–c (Section 16). No integral is evaluated: the Batch A/B/C integrals belong to p15 and later. Opening P.5 entry earns none of the loop or restoration flags. `quantum_inputs_evaluated`, `projected_defect_computed`, `projected_identity_verified` and `projected_restoration_verified` are not earned; `bulk_symmetry_restored`=false. D12 is Missing, never zero. O.1 is NOT_CLOSED. The insertion stays frozen, XV-B and XIV-J are not opened, Route B is not discharged, and no $c\,\delta$ is added to u^R_μ. XVI-P stays *unregistered*: the package is evidence in the sense of XVI-O `conv:package`, not a series certificate, and XVI-A–XVI-O are not edited.

11 P.3/P.4 frozen as P.5 inputs

An independent rebuild of p13 passed. A reviewer unpacked `XVI-P-source-only.zip` (sha256 `2f85a566...`) in a fresh environment and ran the source-only entrypoint. All reference values matched (112/112, no mismatch), 77 values were recomputed by separate code, and a hand-written spot check of the conformal boundary identity agreed. The report is archived as `archival/p13/reproducibility/independent_rebuild_report_p13.json` (sha256 pinned). **Caveat:** the rebuild ran on the same box (shared kernel, TeX Live and poppler) in a fresh virtual environment from the source-only zip; it was not on a second machine.

The flag is *bound to the p13 source tree* `2b1baf33...` and to the p13 zip; p14 changes cannot inherit it, and p14 has its own (pending) independent flag. `P3_P4_frozen_as_P5_inputs` is recomputed in every build by `freeze_p13.py` as the conjunction of three conditions: F1, the archived independent report has its pinned hash, `pass`=true, no mismatch, and is bound to the p13 tree and zip; F2, the 54 frozen files (classical P.3/P.4 code, the inherited XVI-O code and notes, pins and inherited data; `archival/p13/frozen_p13_inputs.json`) are byte-identical; F3, the regenerated classical values (export hashes, the seven classical certificates and two ledgers, residual records, pins) equal `archival/p13/reference_values_p13.json`. The classical code is therefore unchanged in p14. Its certificates keep their `P13_*` names, and the p14 gate recomputes the p13 acceptance from them (Section 19). **P.5 entry is stated against this freeze:** if any of F1–F3 fails, P.5 entry is not earned and publication is rejected.

12 Milestones (p14 status)

milestone	definition (unchanged since p13, `conventions.MILESTONES`)	status (p14)
P.3	fixed output projection + exhaustive coefficient dependencies + graph topology inventory	complete (frozen)
P.4	required classical action, transformations, source terms, gauge map and master verification. **P.4 means classical readiness, NOT coupled one-loop restoration.**	earned (frozen)
P.5 entry	quantum prescription, amplitude rules and counterterm domain fixed	earned iff the five contract flags hold and P.3/P.4 are frozen (see summary)
P.5 calculation	coefficients evaluated and renormalized	open (p15+)
P.5 identity	projected defect computed and, if possible, cancelled	open

P.5 entry := `quantum_action_defined` ∧ `quantum_prescription_defined` ∧ `D11_matching_defined` ∧ `counterterm_domain_defined` ∧ `amplitude_rules_verified`, each recomputed from its certificate, together with `P3_P4_frozen_as_P5_inputs`. The values of this build:

contract item	value (from the certificates of this build)
integrated fields	h_(mn) (10 independent components), c^mu, cbar_mu, phi, b^mu (retained)
external antifields	K'_(mn): h*' (10 comps, sigma_h = +1), c*'_mu: c*', cbar*'^mu: cbar*', phi*': phi*', b*': absent (s b = 0)
b	RETAINED and integrated; det M_bb = -16; eliminating b gives 4 terms quadratic in cbar*'; Gamma_1 independent of bs, cbs
antifield shift	eps = -1 (derived)
prescription	P14_DRED_MSbar_FeynmanI0: MS-bar: subtract the pole in the combination 1/eps - gamma_E + log 4 pi; scale mu
i0	Feynman: every propagator denominator q^2 + i0 (eta = +,-,-,-); Wick rotation l0 = i l4 implied
DRED rules	verified = true; T1 finite a/8 + b/4 (premature d=4 would give b/4); T2 finite -a/2 ((d-4) x pole); mutations: premature_d4 rejected, trace_ghat_eq_4 rejected, contract_after_laurent rejected
D11 matching	defined = true, verified = false; M1: 100 component pairs, 0 mismatches; M3 (conditional on M2): kernel sd <= 8, contact order <= 4; contact space order_0 2, order_2 4, order_4 5; under N4 5 (transverse 2); M2 defined_not_verified
counterterm domain	kappa^1: 0; kappa^2: 11 (h.h 5, c.cb 2, c.K 3, phi.phi 1); sigma_0-equivalent: 3 (h.h 2, c.cb 0, c.K 0, phi.phi 1); 8 candidate directions modulo the computed exact equivalences, before the identity constraints (D cap ker sigma_0 not computed)
amplitude controls	W1_wick_berezin_shadow: pass, W2_multiplicity: pass, W3_closed_form_vertices: pass, W5_regeneration: pass, W6_hessian_route: pass, W4_ward_bose: pass; W5 regenerates D14a-c, D11, D13; W6 (Hessian route) regenerates D01, D02, D03, D12 == exports: true
mutations (must be rejected)	wrong_tadpole_factor: rejected, wrong_ghost_loop_sign: rejected, offdiag_multiplicity: rejected, reversed_source_ghost_order: rejected, omit_ghost_transport_term: rejected
P.5 entry flags	quantum_action_defined = true, quantum_prescription_defined = true, D11_matching_defined = true, counterterm_domain_defined = true, amplitude_rules_verified = true
P3_P4_frozen_as_P5_inputs	true (the p13 independent rebuild ran on the SAME box (shared kernel, TeX Live, poppler) in a fresh venv from the source-only zip; it was NOT run on a second machine)
P.5 entry	earned (contract fixed; stated against the P.3/P.4 freeze)

graph	prefactor	denominators	comps	monomials	deg	status
D01	`kappa**2/2`	l^2+i0, (l-p)^2+i0	100	3888	4	integrand_available
D02	`-kappa**2/2`	l^2+i0	94	220	2	integrand_available
D03	`-kappa**2`	(l-p)^2+i0, l^2+i0	100	2728	4	integrand_available
D11	`kappa**2/2`	l^2+i0, (l-p)^2+i0	100	1722	4	reference
D12	`-kappa**2/2`	l^2+i0	94	160	2	Missing
D13	`I*kappa**2`	l^2+i0, (p-l)^2+i0	40	396	3	integrand_available
D14a	`-kappa/2`	l^2+i0	10	22	2	integrand_available
D14b	`kappa`	l^2+i0	4	16	2	integrand_available
D14c	`-kappa/2`	l^2+i0	10	22	2	integrand_available

13 The quantum object

Builder/output: `quantum_object.py`, `P14_quantum_object.json`.

Action. The action is computed directly in transformed variables:

$$S_\Psi(\Phi,\Phi^{*\prime}) = \sum_k \kappa^{k-2}\int \ell_k + S_\phi + X\big(\Phi,\Phi^{*\prime} + \varepsilon\,\partial\Psi/\partial\Phi\big), \qquad \varepsilon = -1 \text{ (derived)},$$

with $\Psi = -i\int \bar{c}_\mu(2D^\mu[h] - b^\mu)$. By order, X contains $K'c$, $\bar{c}c$, b^2, bh, $b\bar{c}^{*\prime}$ at κ^0; $K'ch$, $\bar{c}ch$, $c^{*\prime}cc$, $\phi^{*\prime}c\phi$ at κ^1; and nothing at κ^2. S_ϕ contributes $\phi\phi$, $h\phi\phi$, $hh\phi\phi$. There are no switch or insertion fields: the baseline is $S_{\rm inv}$, not S_D.

Fields and sources. The integrated fields are $h_{(mn)}$ (10 independent components), c^μ, $\bar{c}_\mu$, ϕ and b^μ. The external antifields are $K'_{(mn)}$ ($\sigma_h = +1$), $c^{*\prime}_\mu$, $\bar{c}^{*\prime\mu}$ and $\phi^{*\prime}$; $b^{*\prime}$ is absent because $sb = 0$.

b **is retained and integrated.** Four reasons:

- b occurs only in κ^0 bilinears, so it has no interaction vertex;
- M_{bb} is constant and invertible (det $= -16$);
- the full (h, b) inverse is verified, with Schur complement q^2P;
- eliminating b (computed) would make S_Ψ *quadratic* in $\bar{c}^{*\prime}$, whereas retaining it keeps S_Ψ linear in every external antifield.

$\bar{c}^{*\prime}$ couples only through $\bar{c}^{*\prime}b$, which has one quantum leg and is tree-level and one-particle reducible. So the 1PI Γ_1 is independent of $\bar{c}^{*\prime}$ and $b^{*\prime}$, while K', $c^{*\prime}$ and $\phi^{*\prime}$ enter through $K'ch$, $c^{*\prime}cc$ and $\phi^{*\prime}c\phi$.

1PI convention. The weight is $e^{iS_\Psi/\hbar}$ and $\Gamma_\Psi = S_\Psi + \hbar\Gamma_1 + O(\hbar^2)$, with the Legendre transform taken in the fields only (antifields are spectators). $\Gamma_1 = \frac{i}{2}\mathrm{STr}\log S^{(2)}_\Psi = -i\log Z|_{\rm 1PI}$, so that $i\Gamma_1$ is the sum of 1PI one-loop graphs (Gate 0; the Berezin/Wick shadow below).

Vacuum and one-point. The vacuum is normalised by $\Gamma_1[0] = 0$. The one-point function (D14a–c) is retained: no tadpole cancellation and no background shift are imposed, and no value is assigned.

Fourier, propagators and i0. We use $\partial \to ik$ with incoming momenta. The propagators are $\langle hh\rangle = i(P^{-1})_{ab}/(q^2 + i0)$ (de Donder; independent components), $\langle c^\mu\bar{c}_\rho\rangle = i(M_0^{-1})_{\mu\rho} = -\delta_{\mu\rho}/(q^2 + i0)$ and $\langle\phi\phi\rangle = i/(q^2 + i0)$, all of them inverses of the verified Hessians; the $i0$ is Feynman throughout.

Bookkeeping. Each n-leg vertex is $O(\kappa^{n-2})$. At one loop, Γ_1 at κ^k has exactly k external legs ($\sum_v(n_v - 2) = E$), and this is checked against the prefactor of every exported integrand.

Pullback, not theorem. $\Gamma_\Psi(\Phi, \Phi^{*\prime}) = \Gamma(\Phi, \Phi^{*\prime} - \partial\Psi/\partial\Phi)$ is used only as the classical pullback (verified at tree level). No quantum gauge-change theorem is asserted.

14 Prescription and the D11 matching note

Builder/output: `prescription_d11.py`, `P14_prescription_D11_matching.json`.

One prescription. Dimensional reduction (DRED), made operational in p14r (Section 14.1). Subtraction is $\overline{\rm MS}$ at scale μ, and the $i0$ is Feynman. The rules for scaleless integrals belong to the definition but are *not applied*: no tadpole (D02, D12, D14a–c) receives a value. D12 stays Missing. XVI-O fixed D12 and the one-point function by `dimensional_regularization_m0`; P's prescription belongs to the same family, but P does not import that value.

14.1 Operational DRED rules (p14r)

Builder/output: `dred.py`, `P14_dred_rules.json`; `quantum_prescription_defined` requires `dred_rules_verified`, and the release gate recomputes it.

R1, spaces. The quasi-4-dimensional space Q4S splits as QdS $\oplus$ Q$(4-d)$S: $g^{(4)} = \hat{g} + \tilde{g}$, with $\mathrm{tr}\, g^{(4)} = 4$, $\mathrm{tr}\,\hat{g} = d$ and $\mathrm{tr}\,\tilde{g} = 4 - d$. The metrics are projectors: $\hat{g}\hat{g} = \hat{g}$, $\tilde{g}\tilde{g} = \tilde{g}$, $\hat{g}\tilde{g} = 0$, $g^{(4)}X = X$.

R2, momenta. The loop momentum l *and* the external momenta p lie in QdS ($p\cdot\tilde{g} = 0$; a stated choice). Field, vertex and polarisation indices are Q4S indices.

R3, lifting. The exported numerators are polynomials in the lower components l_r, p_r ($r = 0..3$) at fixed external components. They are lifted as $l_r := l\cdot e_{(r)} = l\cdot\hat{e}_{(r)}$ and $p_r = p\cdot e_{(r)}$, with $e_{(r)}$ the η-orthonormal basis of Q4S. The vertex tensors are lifted with $\eta \to g^{(4)}$, so nothing is truncated.

R4, reduction. $\int l^{\mu_1}\cdots l^{\mu_n}F(l^2, l\cdot p) = \sum_k C_k B_k$, with B_k the symmetric tensors built from $\hat{g}$ and p, and C_k given by the exact Gram (Passarino–Veltman) projection in symbolic d. Contracting with the lifted basis gives $\hat{g}_{rs} = \eta_{rs} - \tilde{G}_{rs}$, where the evanescent $\tilde{G}_{rs}$ satisfy $\eta^{rs}\tilde{G}_{rs} = 4 - d$ and $\tilde{G}_{rs}p^s = 0$. The checked closed forms are: the rank-2 tadpole $\hat{g}\,J/d$; the rank-4 tadpole $J/(d(d+2))$; $B_{00} = (J_{10}p^2 - J_{02})/((d-1)p^2)$; and $B_{11} = (dJ_{02} - p^2 J_{10})/((d-1)p^4)$.

R5, order. *Every* index contraction, including the external 4D projection, is done *before* the Laurent expansion. d stays symbolic through the $\overline{\text{MS}}$ subtraction and the finite-part extraction, so $(d-4)\times$pole terms are retained.

R6, evanescent. Evanescent remainders *do* arise in general. The lift of the exported integrands has $\tilde{G}$ parts in every graph, and contraction produces $(4-d)\times$pole finite terms. They are kept. The p14 sentence claiming that none arise is withdrawn.

Tests (generic placeholders $J = a/\epsilon + b + c\epsilon$, $d = 4 - 2\epsilon$; no integral of this theory is evaluated).

- T1, the η_{00} part of $\int l_0 l_0 F$, has finite part $a/8 + b/4$. Substituting $d = 4$ before the expansion would give $b/4$; the guarded pipeline raises `PrematureD4Substitution`.
- T2, $\eta^{rs}\times$ the $\tilde{G}$ part, equals $-(4-d)J/d$ with finite part $-a/2$, a $(d-4)\times$pole term.

The mutations $\operatorname{tr}\hat{g} = 4$ (rules check fails; T1 becomes $b/4$ and T2 becomes 0) and expand-before-contract (`UncontractedObjects`) are both rejected. Background literature (nothing is imported from it): W. Siegel, Phys. Lett. B84 (1979) 193 and B94 (1980) 37; D. Stöckinger, JHEP 0503 (2005) 076, hep-ph/0503129; A. Signer and D. Stöckinger, Nucl. Phys. B808 (2009) 88, arXiv:0807.4424.

D11 matching (defined, not verified).

M0, baseline. S_{inv}, not S_D. $a_1 = \frac{1}{2}\tau\eta - h$ is XVI-O's vertex (`lem:d11-prefactor`: $\Pi = (i\kappa^2/4)T$).

M1, exact integrand identity (computed). For all 100 pairs (A, B) of independent components,

$$-i\,\mathcal{P}\,\frac{N^P_{AB}(l,p)}{(l^2+i0)((l-p)^2+i0)} = \frac{i\kappa^2}{4}\,R^{\mu\nu}_A R^{\rho\sigma}_B V^{ab}_{\mu\nu} V^{cd}_{\rho\sigma}\big[\tilde{C}_{ac}(l)\tilde{C}_{bd}(l-p) + \tilde{C}_{ad}(l)\tilde{C}_{bc}(l-p)\big],$$

where:

- $\tilde{C}_{ab}(k) = k_a k_b\, i/(k^2+i0)$ is the Fourier transform of $C = -\partial\partial D_F$;
- V is XVI-O's own `V_tensor`, imported from `code/matter_loop/d11_tensor.py`;
- R_A is the η-raising map $h^{\mu\nu} = \eta^{\mu a}\eta^{\nu b}h_{ab}$, with orbit multiplicity 2 for off-diagonal A;
- $\mathcal{P} = \kappa^2/2$ is P's prefactor, which carries the symmetry factor $\frac{1}{2}$.

The identity matches $\Gamma^{(2)}_1 = \frac{1}{2}\int h^{\mu\nu}\Pi_{\mu\nu\rho\sigma}h^{\rho\sigma}$ (XVI-O) with P's ordered extraction, and it is bound to the exported D11 integrand by its sha256.

M2, nonlocal agreement on $U = \mathbb{R}^4\setminus\{0\}$ (defined only). $\langle \mathcal{F}^{-1}[P^{\text{DR}}_{AB}], f\rangle = \langle R_A R_B \Pi^R_{\text{tr}}, f\rangle$ for $f \in \mathcal{D}(U)$, with $\Pi^R_{\text{tr}} = (i\kappa^2/4)\mathcal{K}u^R_\mu$, $\alpha = A^2/60$ and $\beta = A^2/18$ (`lem:BV_rules_on_U`, all 256 components). This needs the evaluated amplitude, which is p15+ work.

M3, local difference (conditional on M2). $L_{\text{match}} = \mathcal{F}^{-1}[P^{\text{DR}}] - RR\,\Pi^R_{\text{tr}}$. The degrees are as follows. The scalar extension u^R_μ has scaling degree 4. The kernel $\mathcal{K}u^R_\mu$ (with $\mathcal{K}$ of order 4) has scaling degree at most 8. A contact term has $\operatorname{sd}(\partial^\alpha\delta) = 4 + |\alpha|$. *Kernel scaling degree at most eight; degree of divergence at most four; hence the local ambiguity has derivative order at most four.*

If M2 holds, then L_{match} is supported at $\{0\}$ and lies in the bounded contact space of Lorentz-covariant even hh structures of derivative order ≤ 4. This space is enumerated: order 0 (μ^4) has dimension 2 (transverse 0); order 2 ($\mu^2 p^2$) has dimension 4 (transverse 1, the linearised Einstein–Hilbert structure); order 4 has dimension 5 (transverse 2). Odd orders cannot occur with four indices built from p and η.

Exactly degree four is an *explicit normalization condition* N4: no μ^4 and no $\mu^2 p^2$ contact. N4 is *not* implied by "no other scale", since μ is a scale. Under N4 the space is the 5-dimensional order-4 span, equal to the hh sector of the counterterm domain. XVI-O's contact ansatz is a candidate list (`NOT_proved_spanning`) with three named order-4 entries, one of them the collective `L4_mixed`; its lower-order entries are excluded by N4. P's list is the explicit one.

M4, finite adjustments. Under M2 and N4, these are booked only as P-side counterterms in the hh sector (without N4 the order-0/2 contacts would have to be added there). u_μ^R is not modified (no $c\,\delta$), so the scalar-extension prescription `analytic_FP_u_mu_R_o70` is preserved.

M5, residue. $r = -i\pi^2$ is an XVI-O-side statement, valid only *after* the constructive holomorphy of $F(\zeta) = (Q - i0)^\zeta$ (`lem:analytic_FP_u_mu_R_o70`). P's $1/\epsilon$ pole is P-side, and no pole–residue identification is asserted.

M6/M7. The XVI-O o70 files are byte-pinned. XVI-O's kernel Ward identity $\partial(\Pi_w + L) = 0$ is *not* functional ST4b, and the projected identity (XVI-N `lem:field-only`) is distinct from bulk restoration.

Hence `D11_matching_defined`=true and `D11_matching_verified`=false.

15 Counterterm domain

Builder/output: `counterterm_domain.py`, `P14_counterterm_domain.json`.

This is an *order-bounded EFT constraint list, not an admission* (XIV-J and XV-B are not opened). The one-loop κ^n part has n external legs and local dimension $4 + n$, with $[\kappa] = -1$; $[h] = [\phi] = [c] = [\bar{c}] = 1$; and $[K'] = [c^{*\prime}] = [\phi^{*\prime}] = [\bar{c}^{*\prime}] = [b] = 2$. The functionals have ghost number 0, are even under $\phi \to -\phi$ (with $\phi^{*\prime} \to -\phi^{*\prime}$), and are Lorentz covariant (parity-odd structures vanish for two-leg terms). The permitted species are the legs of loop-capable vertices; b, $\bar{c}^{*\prime}$ and $b^{*\prime}$ are excluded. The structures are *generated* mechanically from η and the leg momentum and turned into densities, and their ranks are exact. Total derivatives vanish identically in the two-point representation.

- κ^1: empty (a one-leg term with derivatives is a total derivative, and there is no dimension-5 field).
- κ^2: 11 structures.
  - hh, 5: $h\Box^2 h$, $\tau\Box^2\tau$, $(\partial\partial h)\Box\tau$, $(\partial h)\Box(\partial h)$, $(\partial\partial h)^2$;
  - $\bar{c}c$, 2: $\bar{c}_\mu\Box^2 c^\mu$, $\bar{c}_\mu\partial^\mu\Box\partial_\nu c^\nu$;
  - $K'c$, 3: $K'^{mn}\partial_m\partial_n\partial_r c^r$, $K'^m{}_m\Box\partial_r c^r$, $K'^{mn}\Box\partial_m c_n$;
  - $\phi\phi$, 1: $\phi\Box^2\phi$.

Source dependence. K' enters only via $K'c$. $c^{*\prime}$ and $\phi^{*\prime}$ appear first in three-leg terms at κ^3, outside the order bound.

Equivalences. Apply $\sigma_0 = (S_\Psi^{(0)}, \cdot)$ (frozen antibracket) to every generated ghost-number -1, dimension-5 bilinear Ξ. Antifield-linear Ξ are field redefinitions (canonical transformations); field-only Ξ are s-exact. The domain $\cap\,\mathrm{Im}\,\sigma_0$ is given in the summary; the $S_\Psi^{(0)}$ density used for this is checked against the verified Hessians. *Eight candidate directions modulo the computed exact equivalences, before imposing the relevant identity constraints.* This is not a cohomological classification, which needs $D \cap \ker\sigma_0$ (the σ_0-closedness constraints) and the quotient $(D \cap \ker\sigma_0)/(D \cap \mathrm{Im}\,\sigma_0)$. $D \cap \ker\sigma_0$ has not been computed, so the count 8 is an upper bound only. The enumeration itself is unchanged.

Normalisation. $\overline{\mathrm{MS}}$ (pure poles at μ). There is no κ^2 two-derivative counterterm, so neither κ nor the h normalisation is renormalised at this order. This is the *imposed* normalization condition N4 of M3, not a consequence of masslessness. There is no κ^1 counterterm. Finite adjustments stay inside the domain.

16 Executable amplitude rules

Builder/output: `feynman.py`, `shadow.py`, `amplitude_rules.py`, `P14_amplitude_rules.json`, `exports/P14_vertex_tensors.json`, `exports/P14_integrands/*.json`.

Rules.

- *Extraction:* ordered, $V[L_1..L_n] = \partial^L_{L_n} \cdots \partial^L_{L_1} S$, with leg 1 leftmost; the Feynman vertex is $i\kappa^n V$.
- *Graviton legs:* independent components; the multiplicity (1 diagonal, 2 off-diagonal) enters only the tensor form.
- *Ghost:* $\bar{c}$ is on the left in $\bar{c}ch$ and in M_0; a closed ghost loop carries -1.
- *Source:* K' sits left of c in $K'ch$ and in the Γ_1 coefficient $K'_{a'} G_{a'\nu} c^\nu$.
- *Symmetry factors:* $S_f = \frac{1}{2}$ for D01, D02, D11, D12, D14a, D14c and $S_f = 1$ for D03, D13, D14b.
- *Ghost-loop sign:* -1 for D03 and D14b.

The vertex tensors include every derivative term of the ghost-graviton vertex ($\bar{c}\,c\,h = 2ia\,D^\rho[\mathcal{L}_c h]$, including the transport term $c^r \partial_r h$) and the K' vertex.

Controls.

W1. Berezin/Wick shadow theory: the symmetry factors and Grassmann signs are fixed independently.

W2. A tensor-route recomputation of the multiplicities.

W3. Closed forms of the vertices.

W4. The primary evidence is the generic symbolic *off-shell* Ward identities at κ^1 and κ^2. A supplementary check is the "Ward contraction with two free on-shell graviton legs and an off-shell gauge leg". W4 also checks Bose symmetry.

W5. Independent sympy regeneration of D14a, D14b, D14c, D11 and D13 *only*, plus a Bose-symmetry check of D01.

W6 (p14r). Independent regeneration of D01, D02, D03 and D12 from the one-loop expansion of the functional (super)trace log of the field-dependent Hessian of S_Ψ (`hessian_route.py`), not from the Feynman-rule assembler:

$$i\Gamma^{AB} = -\tfrac{1}{2}\,\mathrm{Tr}(GU_2^{AB}) + \tfrac{1}{2}\,\mathrm{Tr}(GU_1^A GU_1^B) \text{ (bosons)}, \qquad i\Gamma^{AB} = -\,\mathrm{Tr}(GU_1^A GU_1^B) \text{ (ghosts)}.$$

The ingredients are an own monomial differentiation of ℓ_3, ℓ_4, an own inverse of the full 14×14 (h, b) Hessian, and the ghost and scalar Hessians from the S_Ψ densities. Structural checks: $X^{(1)}$ has no field-only term and $X^{(2)} = 0$, so there is no $\bar{c}chh$ vertex. The route equals the exported integrands exactly, component by component (D01 100, D02 94, D03 100, D12 94; 0 mismatches), and it also rejects the tadpole-factor, ghost-sign, multiplicity and transport-term mutations.

Mutations that must be rejected (all rejected): wrong tadpole factor, wrong ghost-loop sign, off-diagonal multiplicity, reversed source/ghost order, and omitted ghost transport term. D13 has priority; its integrand is exported with prefactor $i\kappa^2$ and order (K', c). Integrands are symbolic (not integrated); the per-graph table is in the summary above.

Isolated-copy gate tests (p14r). The test copies contain every frozen input (the top-level directories of `frozen_p13_inputs.json`, including `tests/`); the p14 copies lacked `tests/`, so every copy was unfrozen. The expected outcomes are:

- the unmutated copy has freeze true, and publication is accepted;
- each P.5-only mutation (DRED mutation not rejected, counterterm flag, amplitude mutation) keeps the freeze *true*, turns the named contract flag false, and is rejected;
- the hard-stop mutations keep the freeze true and are rejected;
- the classical mutations make the freeze false and are rejected.

17 P.5 entry dossier

One row per claim, each resolved mechanically by `dossier_p14.py`: the function exists, the certificate key holds the expected value, and the test exists. The five P.3/P.4 chains are carried over unchanged as frozen inputs; the P.5 entry chain is new.

id	claim	file :: function	certificate key = value	test
P.3: projection -> dependency rows -> graphs [frozen P.3/P.4 input]				
P3.1	ch output projection fixed (c^1 h^1, ghost number 1, kappa^2, one loop, transformed variables)	ch_source_audit.py :: main	ch_source_audit/ch_projection_defined = true ch_source_audit/ch_pattern/ghost_number = 1	ch_rows_and_quantum_input_slots
P3.2	vertex inventory generated from S_Psi (not copied)	ch_source_audit.py :: S_psi_inventory, vertex_options	ch_source_audit/vertex_inventory = {"kappa0": ["K.c",... ch_source_audit/checks/b_has_no_interaction_vertex = true	ch_rows_and_quantum_input_slots
P3.3	dependency rows exhaustive: regenerated rows equal the published rows	ch_source_audit.py :: audit, regenerate_and_compare	ch_source_audit/checks/regenerated_rows_match_published = true ch_source_audit/regeneration/complete = true	e2e_dropped_hstar_c_row
P3.4	6 rows: 3 surviving, 3 derived zero (S2, cbar*, b* rows)	ch_source_audit.py :: audit	ch_source_audit/rows_total = 6 ch_source_audit/rows_surviving = 3 ch_source_audit/rows_derived_zero = 3	ch_rows_and_quantum_input_slots
P3.5	enumeration lemma (justifies max_len = 2) checked mechanically	ch_source_audit.py :: enumeration_lemma	ch_source_audit/enumeration_lemma/holds = true	enumeration_lemma
P3.6	graph topologies generated and named; no unnamed graph	ch_source_audit.py :: enumerate_graphs, name_graph	graph_inventory/names_TBD = [] graph_inventory/graphs = {"D03_hh_ghost_loo... ch_source_audit/checks/names_TBD_empty = true	ch_rows_and_quantum_input_slots
P3.7	P.3 flags	release_p14.py :: classical_block	RELEASE_STATUS/classical/flags/ch_projection_defined = true RELEASE_STATUS/classical/flags/ch_dependency_inventory_complete = true	milestones_consistent
P.4: classical coefficient -> export -> verification [frozen P.3/P.4 input]				
P4.A1	C3 boundary identity sqrt(-g)R - L_GG = -d_mu B^mu on generic jets	boundary_identity.py :: generic_jet_proof	boundary_identity/C3_proved = true	C3_boundary_identity_generic_and_conformal
P4.A2	ell_2, ell_3, ell_4 built and exported; export hash == quartic cert == master input	quartic.py :: build_and_export	RELEASE_STATUS/classical/hash_consistency/consistent = true quartic_verified/ell4_sha256 = "33ac9e49c4d1644ee...	stale_ell4_hash_blocks_master
P4.A3	C1 inverse, C2 det/trace-log (10 symbolic components)	quartic.py :: inverse_control, det_tracelog_generic	quartic_verified/acceptance_controls_deterministic/C1 = true quartic_verified/acceptance_controls_deterministic/C2 = true	quartic_acceptance_all_deterministic
P4.A4	C4 export == independent evaluator (exact polynomial identity)	quartic.py :: export_identity	quartic_verified/acceptance_controls_deterministic/C4 = true	quartic_acceptance_all_deterministic
P4.A5	C5/C6 ell_3 = XII-C, ell_2 = XI-A Fierz-Pauli modulo divergences (pinned XVI-N grav_density)	quartic.py :: regression_deterministic, xvin_symbolic	quartic_verified/acceptance_controls_deterministic/C5 = true quartic_verified/acceptance_controls_deterministic/C6 = true	quartic_acceptance_all_deterministic
P4.A6	C7 gravitational Noether identities on the exports, kappa^0..kappa^2	grav_noether.py :: noether	quartic_verified/controls/C7_accepted_orders = *all entries true*	grav_noether_exact_and_mutation
P4.A7	quartic mutations rejected; quartic flag	quartic.py :: main	quartic_verified/mutations_all_rejected = true RELEASE_STATUS/classical/flags/quartic_generator_verified = true	e2e_corrupted_ell4_coefficient
P.4: transformations -> source terms [frozen P.3/P.4 input]				
P4.B1	ghost law derived from s^2 = 0 (unique lambda = 1)	brst_from_geometry.py :: solve_ghost_law	brst_derivation/ghost_law_derivation/lam_solutions = ["1"]	brst_and_source_terms_derived
P4.B2	s_2 derived zero in every sector; s nilpotent at all orders	brst_from_geometry.py :: per_sector_expansion, nilpotency	brst_derivation/s2_per_sector = {"h": "derived_zer... brst_derivation/nilpotency_exact_all_orders = true	brst_and_source_terms_derived
P4.B3	sigma_A derived from the generator rule (unique); antifield-linear, no kappa^2 X terms	source_action.py :: derive_sigma, build_X_inv	source_action/sigma_unique = true source_action/X_kappa2_terms = 0 source_action/antifield_linear = true	brst_and_source_terms_derived e2e_flipped_odd_sign e2e_omitted_cstar_term
P.4: gauge map -> Hessians -> source convention [frozen P.3/P.4 input]				
P4.C1	antifield shift orientation derived (eps = -1); S_Psi = S + s Psi mod d	gauge_fixing_map.py :: build, shift_map	gauge_fixing_map/derived_eps = -1 gauge_fixing_map/S_Psi_equals_S_plus_sPsi_mod_d = true	e2e_altered_gauge_shift

id	claim	file :: function	certificate key = value	test
P4.C2	canonicity of the map on all 299 (anti)field pairs	gauge_fixing_map.py :: canonicity	gauge_fixing_map/canonicity/pass = true gauge_fixing_map/canonicity/pairs_tested = 299	gauge_hessian_generic_and_full_canonicity
P4.C3	graviton Hessian at generic symbolic k: Schur complement k^2 P, propagator (k^2 P)^-1	gauge_fixing_map.py :: hessian_check	gauge_fixing_map/propagators/graviton/schur_equals_k2P = true gauge_fixing_map/propagators/graviton/propagator_hh_equals_inv_k2P = true	gauge_hessian_generic_and_full_canonicity
P4.C4	ghost (i cbar Box c) and scalar (-Box) kinetic terms; propagators match transformed Hessians	gauge_fixing_map.py :: ghost_kinetic_check	gauge_fixing_map/propagators/ghost/match = true gauge_fixing_map/propagators/scalar/match = true gauge_fixing_map/propagators_match_transformed_Hessians = true	gauge_hessian_generic_and_full_canonicity
P4.C5	source convention Gamma_Psi(Phi,Phi*') := Gamma(Phi, Phi*' - dPsi/d-Phi): classical pullback, NOT a quantum theorem	gauge_fixing_map.py :: build	gauge_fixing_map/source_identity_relation_defined = true gauge_fixing_map/source_identity_relation/status = "DEFINED classical...	gauge_hessian_generic_and_full_canonicity
P4.C6	gauge-map flag	release_p14.py :: classical_block	RELEASE_STATUS/classical/flags/gauge_fixing_map_constructed = true	milestones_consistent
P.4: master component -> residual record -> acceptance gate [frozen P.3/P.4 input]				
P4.D1	master equation at kappa^0: every antifield-number component; grav af0 record; gate	master_kappa2.py :: run, XX_by_order, grav_deterministic	master_kappa2/table_by_kappa_and_antifield_number/kappa0 = *all entries true* master_kappa2/details/grav_af0_deterministic/kappa0/Q_terms = [0, 0, 0, 0] master_kappa2/details/grav_af0_deterministic/kappa0/Q_identically_zero = true RELEASE_STATUS/classical/residual_records/master_grav_per_order/kappa0/accepted = true	residual_records_regenerated_and_agree
P4.D2	master equation at kappa^1: every antifield-number component; grav af0 record; gate	master_kappa2.py :: run, XX_by_order, grav_deterministic	master_kappa2/table_by_kappa_and_antifield_number/kappa1 = *all entries true* master_kappa2/details/grav_af0_deterministic/kappa1/Q_terms = [0, 0, 0, 0] master_kappa2/details/grav_af0_deterministic/kappa1/Q_identically_zero = true RELEASE_STATUS/classical/residual_records/master_grav_per_order/kappa1/accepted = true	residual_records_regenerated_and_agree
P4.D3	master equation at kappa^2: every antifield-number component; grav af0 record; gate	master_kappa2.py :: run, XX_by_order, grav_deterministic	master_kappa2/table_by_kappa_and_antifield_number/kappa2 = *all entries true* master_kappa2/details/grav_af0_deterministic/kappa2/Q_terms = [0, 0, 0, 0] master_kappa2/details/grav_af0_deterministic/kappa2/Q_identically_zero = true RELEASE_STATUS/classical/residual_records/master_grav_per_order/kappa2/accepted = true	residual_records_regenerated_and_agree
P4.D4	no required master residual; no unconditional pass field	master_kappa2.py :: master_residuals	master_kappa2/required_master_residuals = []	master_no_unconditional_pass_and_deterministic
P4.D5	count/flag invariant on every record; quartic and master records agree	certio.py :: invariant_violations, cross_check	RELEASE_STATUS/classical/deterministic_prerequisites/residual_record_invariants_all_certificates = true RELEASE_STATUS/classical/residual_records/cross_certificate/agree = true	certificate_mutation_rejects_publication certio_rejects_inconsistent_record_before_serialization
P4.D6	acceptance gate: release_claim_accepted and classical readiness (P.4)	release_p14.py :: classical_block	RELEASE_STATUS/classical/release_claim_accepted = true RELEASE_STATUS/classical/classical_readiness_for_ch_identity = true RELEASE_STATUS/classical/flags/required_classical_BV_components_verified = true	release_accepted_and_hard_stops
P4.D7	publication gate re-invoked before every pdf/zip/manifest stage, also on resumed builds	build.py :: publication_gate, pdf_readback	(build-time gate)	resumed_stage_build_blocked
P.5_entry: quantum object -> prescription/D11 matching -> counterterm domain -> amplitude rules -> gate				

id	claim	file :: function	certificate key = value	test
P5E.1	quantum object in transformed variables: S_Psi parts, eps = -1, b retained (elimination makes S quadratic in cbar*'), Gamma_1 independent of cbar*', b*'; pullback != quantum theorem	`quantum_object.py :: main, patterns, b_elimination`	`quantum_object/quantum_action_defined` = true `quantum_object/Q1_transformed_variables/eps_derived` = -1 `quantum_object/Q2_fields_antifields_b/b_decision` = "RETAINED and integrated" `quantum_object/Q2_fields_antifields_b/Gamma1_independent_of` = ["bs", "cbs"] `quantum_object/Q5_kappa_loop_bookkeeping/ok` = true `quantum_object/Q6_pullback_vs_theorem/quantum_gauge_change_theorem_asserted` = false	`quantum_object`
P5E.2	ONE prescription: operational DRED (rules R1-R6), MS-bar, Feynman i0; scaleless rules defined but not applied; quantum_prescription_defined requires the verified DRED rules	`prescription_d11.py :: main`	`prescription_D11_matching/quantum_prescription_defined` = true `prescription_D11_matching/prescription/id` = "DRED_MSbar_FeynmanI0" `prescription_D11_matching/prescription_checks` = *all entries true* `prescription_D11_matching/prescription_checks/operational_DRED_rules_verified` = true `prescription_D11_matching/D12_status` = "Missing"	`prescription_and_D11_matching`

P5E.2b	operational DRED: g4 = ghat + gtilde (tr d, 4-d), l and p in QdS, l_r = l.e_(r) lifting, ghat-projected Gram/PV reduction, contractions before Laurent with d symbolic; evanescent parts retained; premature d=4, tr ghat=4 and expand-before-contract are rejected	`dred.py :: reduce_tensor, eta_trace_Gt, toy_four_d_part, toy_evanescent_trace, known_reductions, lift_exported, main`	`dred_rules/dred_rules_verified` = true
			`dred_rules/mutations_all_rejected` = true
			`dred_rules/mutations/premature_d4/rejected` = true
			`dred_rules/mutations/trace_ghat_eq_4/rejected` = true
			`dred_rules/mutations/contract_after_laurent/rejected` = true
			`dred_rules/toy_checks/T1_four_d_part/finite` = "a/8 + b/4"
			`dred_rules/toy_checks/T2_evanescent_trace/finite` = "-a/2"
			`dred_rules/known_reductions/pass` = true
			`dred_rules/evanescent_parts_present_in_exports` = true
			`dred_rules/loop_evaluations_performed` = false
				`dred_rules_certificate`
				`dred_premature_d4_changes_finite_part_and_is_caught`
				`dred_evanescent_trace_is_d_minus_4_times_pole`
				`dred_trace_ghat_eq_4_rejected`
				`dred_known_reductions_and_contraction_engine`

id	claim	file :: function	certificate key = value	test
P5E.3	D11 matching DEFINED (M1 exact integrand identity vs XVI-O V_tensor, 100 pairs; M3 CONDITIONAL on M2: sd(u^R) 4, kernel sd <= 8, contact derivative order <= 4, contact space orders 0/2/4 = 2/4/5, degree 4 imposed by normalization condition N4; no c delta; r = -i pi^2 XVI-O side only); NOT verified	prescription_d11.py :: M1_identity, M3_local_space, M3_contact_analysis	prescription_D11_matching/D11_matching_defined = true prescription_D11_matching/D11_matching_verified = false prescription_D11_matching/matching/M1_integrand_identity/mismatches = [] prescription_D11_matching/matching/M1_integrand_identity/component_pairs_checked = 100 prescription_D11_matching/matching/M1_integrand_identity/bound_to_export = true prescription_D11_matching/matching/M3_local_difference/labels/scalar_extension_scaling_degree = 4 prescription_D11_matching/matching/M3_local_difference/labels/kernel_scaling_degree_bound = 8 prescription_D11_matching/matching/M3_local_difference/labels/local_contact_derivative_order_bound = 4 prescription_D11_matching/matching/M3_local_difference/lower_degree_contacts_excluded_by_N4/order_2_mu2 = 4 prescription_D11_matching/matching/M3_local_difference/dimension = 5 prescription_D11_matching/matching/M3_local_difference/transverse_subspace_dimension = 2 prescription_D11_matching/matching/M4_finite_adjustments/u_R_c_delta_added_by_P = false prescription_D11_matching/matching/M5_residue/r = "-i * pi**2"	prescription_and_D11_matching
P5E.4	counterterm domain: kappa^1 empty, kappa^2 basis 11 (hh 5, cbar c 2, K'c 3, phi phi 1) generated mechanically; eight candidate directions modulo the computed exact equivalences, before imposing the identity constraints (no cohomological classification: D cap ker sigma_0 not computed); constraint list, not an admission	counterterm_domain.py :: structures, admissible_pairs, rank, main	counterterm_domain/counterterm_domain_defined = true counterterm_domain/kappa1/basis_size = 0 counterterm_domain/kappa2/basis_size = 11 counterterm_domain/equivalences/quotient_dimension = 8 counterterm_domain/equivalences/D_cap_ker_sigma0_computed = false counterterm_domain/equivalences/S0_density_check/ok = true counterterm_domain/XIV_J_opened = false counterterm_domain/XV_B_opened = false	counterterm_domain
P5E.5	amplitude rules: vertices (incl. ghost-graviton derivative terms, K' vertex), propagators, ordering, multiplicities, symmetry factors, signs; symbolic integrands; W1-W6 controls (W5 regenerates only D14a-c, D11, D13; W6 regenerates D01, D02, D03, D12)	amplitude_rules.py :: build_integrands, verify, W2, W3, W4, W5	amplitude_rules/amplitude_rules_verified = true amplitude_rules/controls/W1_wick_berezin_shadow/pass = true amplitude_rules/controls/W4_ward_bose/pass = true amplitude_rules/integrands_only = true amplitude_rules/loop_evaluations_performed = false amplitude_rules/graphs/D12/D12_status = "Missing"	amplitude_rules_and_mutations
P5E.5b	independent Hessian/Tr-log regeneration of D01, D02, D03, D12 from the field-dependent Hessian of S_Psi (own (h,b) inverse; not the Feynman-rule assembler) == exported integrands	hessian_route.py :: ell_table, graviton_inverse, ghost_inverse, reference, W6, compare_exports	amplitude_rules/controls/W6_hessian_route/pass = true amplitude_rules/W6_exports/all_agree = true amplitude_rules/W6_exports/D01/n_mismatches = 0 amplitude_rules/W6_exports/D03/n_mismatches = 0	hessian_route_regenerates_D01_D02_D03_D12
P5E.6	mutations rejected: tadpole factor, ghost-loop sign, off-diagonal multiplicity, source/ghost order, ghost transport term	amplitude_rules.py :: apply_mutations, main	amplitude_rules/mutations_all_rejected = true amplitude_rules/mutations/omit_ghost_transport_term/rejected = true amplitude_rules/mutations/reversed_source_ghost_order/rejected = true	amplitude_rules_and_mutations
P5E.7	Berezin/Wick shadow fixes symmetry factors and Grassmann signs independently	shadow.py :: expect, gamma1, run_shadows	amplitude_rules/controls/W1_wick_berezin_shadow/pass = true	amplitude_rules_and_mutations

id	claim	file :: function	certificate key = value	test
P5E.8	P.3/P.4 freeze bound to the p13 tree (independent rebuild passed; same-box caveat); frozen code and values unchanged	`freeze_p13.py` :: `freeze`, `F1_independent`, `F2_frozen_code`, `F3_frozen_values`	`RELEASE_STATUS/P3_P4_frozen_as_P5_inputs` = true `RELEASE_STATUS/p13_independent_rebuild/bound_to_source_tree_sha256` = "2b1baf335468944aa... `RELEASE_STATUS/p13_independent_rebuild/caveat` = "the p13 independe... `RELEASE_STATUS/freeze/F2/changed_or_missing` = [] `RELEASE_STATUS/freeze/F3/differences` = []	`freeze_binding`
P5E.9	P.5 entry gate: earned iff all five contract flags hold; loop/restoration flags NOT earned	`release_p14.py` :: `_main`, `p5_entry_flags`, `milestones_p14`	`RELEASE_STATUS/P5_entry_flags` = *all entries true* `RELEASE_STATUS/P5_entry_earned` = true `RELEASE_STATUS/hard_stops_hold` = true `RELEASE_STATUS/hard_stops/quantum_inputs_evaluated` = "not_earned" `RELEASE_STATUS/hard_stops/projected_identity_verified` = "not_earned" `RELEASE_STATUS/hard_stops/bulk_symmetry_restored` = false `RELEASE_STATUS/D11_matching_verified` = false `RELEASE_STATUS/D12_status` = "Missing" `RELEASE_STATUS/hard_stops/O1_closure` = "NOT_CLOSED"	`p5_entry_gate_and_hard_stops`
P5E.10	publication gate (p14) re-invoked before pdf/zip/manifest stages	`build.py` :: `publication_gate`, `pdf_readback`	(build-time gate)	`resumed_stage_build_blocked`
Quantum input slots (named dependencies, NOT evaluated coefficients)				
slot	hh at kappa^2 (Γ_1 pattern `h^2`)	`ch_source_audit.py` :: `audit`	`ch_source_audit/surviving_rows/0` = D01, D02, D03, D11, D12	`ch_rows_and_quantum_input_slots`
slot	K'c at kappa^2 (Γ_1 pattern `K.c`)	`ch_source_audit.py` :: `audit`	`ch_source_audit/surviving_rows/1` = D13	`ch_rows_and_quantum_input_slots`
slot	h one-point at kappa (Γ_1 pattern `h`)	`ch_source_audit.py` :: `audit`	`ch_source_audit/surviving_rows/2` = D14a, D14b, D14c	`ch_rows_and_quantum_input_slots`
Reproducibility flags (package evidence, read from RELEASE_STATUS at build time; p14 flags bound to the p14 tree): `author_rebuild_passed` = false; `independent_rebuild_passed` = false; `P3_P4_frozen_as_P5_inputs` = true; p13 `independent_rebuild_passed` = true (bound to the p13 source tree `2b1baf335468944a...`; caveat: the p13 independent rebuild ran on the SAME box (shared kernel, TeX Live, poppler) in a fresh venv from the source-only zip; it was NOT run on a second machine).				

18 Reproducibility contract (p14r)

- `author_rebuild_passed` is bound to the *p14r* source tree (`reproducibility/author_rebuild_report_p14r.json`).
- The p14r `independent_rebuild_passed` is `pending`.
- The p13 independent result is recorded separately and bound to the p13 tree; this is the same-box caveat above.
- The source-only archive is `XVI-P-p14r-source-only.zip`, with reference values in `reproducibility/reference_values_p14r.json` and entrypoint `python3 tools/rebuild_from_source.py`.
- The p13 source-only zip, its reference values and the independent report are archived unchanged in `archival/p13/`. The p14 release (zip, reference values, author report) is archived in `archival/p14/`.

Non-claim 1. No loop is evaluated, no one-loop coefficient is fixed, and no symmetry is restored. P.5 entry fixes a contract: an action, a prescription, a matching *definition*, a counterterm *domain* and integrands. P.5 calculation and P.5 identity are open. The prepared S_D keeps $(S_D, S_D) \neq 0$ (XVI-N prop:broken-square), and no completion contradicting XVI-N thm:massless-obstruction is claimed. En01, Em01, Em02-B/D, En02, VII-E and VIII-B are not premises.

19 Frozen classical record (P.3/P.4, unchanged since p13)

Frozen P.5 inputs.
The sections below are the p13 classical record, carried over verbatim (only the release-gate and test file names are updated). The code that produces it is byte-identical to the p13 source tree `2b1baf33...`; its certificates keep the names `P13_*` and are regenerated in every p14 build. The regenerated values are compared with `archival/p13/reference_values_p13.json` (`freeze_p13.F3_frozen_values`).

20 Conventions

Declared once in `conventions.py`.

$\eta = \mathrm{diag}(+,-,-,-)$; $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ (affine in h); $S_{\rm EH} = -(2/\kappa^2)\int \sqrt{-g}R = \sum_k \kappa^{k-2}\int \ell_k$ with $\ell_k = -2[\varepsilon^k]L_{\Gamma\Gamma}$, $L_{\Gamma\Gamma} = \sqrt{-g}\, g^{\mu\nu}(\Gamma^\lambda_{\mu\sigma}\Gamma^\sigma_{\nu\lambda} - \Gamma^\lambda_{\mu\nu}\Gamma^\sigma_{\lambda\sigma})$; $R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma\Gamma - \Gamma\Gamma$, $R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}$. Massless scalar $S_\phi = \frac{1}{2}\int \sqrt{-g}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ with $\sqrt{-g}g^{\mu\nu} = \eta + \kappa a_1 + \kappa^2 a_2$ (XI-A, re-derived in `P13_source_action`). Ghost scaling $\xi^\mu = \kappa a c^\mu$, $a = \frac{1}{2}$. Antibracket (XVI-N def:antibracket)

$$(F,G) = \sum_A \int \Big[\frac{\delta^R F}{\delta\Phi^A}\frac{\delta^L G}{\delta\Phi^*_A} - \frac{\delta^R F}{\delta\Phi^*_A}\frac{\delta^L G}{\delta\Phi^A}\Big], \qquad s\Phi = (S,\Phi),\ s\Phi^* = (S,\Phi^*).$$

Canonical pair for the graviton: (h_{mn}, K^{mn}), $m \le n$. Divergence test (exact): a local density without constant term is a total divergence iff all its Euler derivatives vanish (contractible jet space).

21 C3 repaired: the boundary identity and its orientation

Builder/output: `boundary_identity.py`, `P13_boundary_identity.json`.

Defect. The p09/p10 control C3 evaluated the identity on $g = \eta + A(x^0)\, e_{00}$. This metric is locally flat ($x^0 \mapsto \int\sqrt{1+A}\,dx^0$ removes A); its only nonzero symbol is $\Gamma^0{}_{00}$, and $\sqrt{-g}R = L_{\Gamma\Gamma} = B^\mu = \partial_\mu B^\mu = 0$ identically. The check was therefore vacuous and could not detect a sign; the source comment "keeps ∂B nontrivial" was false. p09/p10 also stated the orientation as $\sqrt{-g}R - L_{\Gamma\Gamma} = +\partial_\mu B^\mu$. That orientation is wrong.

Proposition 2 (Boundary identity, corrected orientation)**.** *With* $B^\mu = \sqrt{-g}\,(g^{\mu\nu}\Gamma^\lambda_{\nu\lambda} - g^{\lambda\nu}\Gamma^\mu_{\lambda\nu})$,

$$\sqrt{-g}\,R - L_{\Gamma\Gamma} = -\,\partial_\mu B^\mu.$$

Deterministic proof (three parts). (i) *Orientation.* The statement above, with $-\partial_\mu B^\mu$, replaces the p09/p10 statement; it is used in `conventions.BOUNDARY_IDENTITY`, the exports and all certificates. (ii) *Nontrivial conformal control.* For $g = f\eta$ with f a generic function of all four coordinates, sympy gives exactly $\sqrt{-g}R = -3\Box f + \frac{3}{2f}(\partial f)^2$, $L_{\Gamma\Gamma} = \frac{3}{2f}(\partial f)^2$, $B^\mu = 3\partial^\mu f$. The identity holds with residual 0; the p10 orientation leaves $-6\Box f \neq 0$. (iii) *Generic jet proof.* Let G^{ab} be ten *free* symmetric symbols (the inverse metric), and let the 40 first jets $g_{ab,c}$ and 100 second jets $g_{ab,cd}$ be free. Use $\partial_c G^{ab} = -G^{ar}g_{rs,c}G^{sb}$ and $\partial_c\sqrt{-g} = \frac{1}{2}\sqrt{-g}\,G^{ab}g_{ab,c}$ (Jacobi; both checked exactly via $\partial\det = \mathrm{tr}(\mathrm{adj}\,\partial g)$ and $\mathrm{adj}\,g = \det g\,\mathbf{1}$). Then $(\sqrt{-g}R - L_{\Gamma\Gamma} + \partial_\mu B^\mu)/\sqrt{-g}$ is a polynomial in 150 variables over $\mathbb{Q}$ (2406, 1764 and 2442 terms in the three pieces) and is *identically zero* for all ten components, first and second jets; with the p10 orientation it has 2736 nonzero terms. Because the identity holds for free G, it holds a fortiori for $G = g^{-1}$. □

Integrated-action equivalence is unaffected. For compactly supported h, $\int\partial_\mu B^\mu = 0$ with either sign, so $S_{\rm EH} = -(2/\kappa^2)\int L_{\Gamma\Gamma}$ and $\ell_k = -2[\varepsilon^k]L_{\Gamma\Gamma}$ are unchanged. The error concerned the stated local identity, not the action.

22 Quartic generator: deterministic acceptance

Builder/output: `quartic.py`, `grav_noether.py`, `series_eval.py`, `jetpoly.py`; `P13_quartic_verified.json`.

ℓ_2, ℓ_3, ℓ_4 are built from $L_{\Gamma\Gamma}$ and exported (`exports/ell{2,3,4}_generating_polynomial.srepr`; ℓ_4 sha256 `33ac9e49...e123`, unchanged from p10; 3360 monomials, a reproducibility count only). *Acceptance* consists only of exact symbolic identities:

control	identity (exact)	status
C1	$q\,q^{-1} = 1 + O(\varepsilon^5)$, all 10 symbolic components	deterministic, holds
C2	$\log(-\det q) = \mathrm{tr}\log(\eta^{-1}q)$, $[\sqrt{\cdot}]^2_{\mathrm{series}} = -\det q$ to ε^4, 10 comp.	deterministic, holds
C3	$\sqrt{-g}R - L_{\Gamma\Gamma} = -\partial_\mu B^\mu$ (generic jets + conformal control)	deterministic, holds
C4	export ℓ_k = independent evaluator in $\mathbb{Q}$[50 jet symbols], $k = 2,3,4$	polynomial identity
C5	$E_{h_{mn}}(\ell_3 - \ell_3^{\text{XII-C}}) = 0$, all 10 components	deterministic, holds
C6	$E_{h_{mn}}(\ell_2 - \ell_2^{\text{FP}}) = 0$, all 10 components	deterministic, holds
C7	$E_{c^\mu}(s_0\ell_{n+2} + s_1\ell_{n+1}) = 0$, $n = 0,1,2$, on the exports	deterministic, holds
reg.	numeric evaluator at random jets; plane-wave cubic regression	*sampled regression control only*

Identification of the exported coefficients. The independent evaluator computes $L_{\Gamma\Gamma}(\eta + \varepsilon h)$ from its definition with the series inverse (C1) and the series $\sqrt{-q}$ (C2), truncated at ε^4 (exact for $k \le 4$). C4 proves that the exports equal $-2[\varepsilon^k]L_{\Gamma\Gamma}$ as polynomials, so no sampling is involved. C3 gives $L_{\Gamma\Gamma} = \sqrt{-g}R + \partial_\mu B^\mu$, hence $\int \ell_k = [\kappa^{k-2}]S_{\mathrm{EH}}$ for compact support. C5 compares, with the same sign and normalisation, against XVI-N `grav_density.ell3_and_curvature` (XII-C $\ell_3 = -2(r_3 + b_1 r_2 + b_2 r_1)$), evaluated on *symbolic* second jets; the opposite sign is not a divergence. This is not a claim that XII-C is the quartic, and not a Ward identity. C7 is exact because W_n is linear in c: $W_n \simeq c^\mu E_{c^\mu}(W_n)$, so W_n is a divergence iff $E_{c^\mu}(W_n) \equiv 0$. At κ^1 and κ^2 the s_0 piece alone leaves 159 and 2088 terms per component, so the cancellation is genuinely mixed. Mutations (each rejected by a deterministic test): M1 one ℓ_4 coefficient $\times\frac{7}{5}$ and M2 sign of ℓ_4 (C7 at κ^2); M3 swapped contraction (C4 for all k, C5, C6, C7 at all orders); M4 missing ε^2 coefficient of $\sqrt{-q}$ (C4 for ℓ_4; C7 at κ^2); M5 wrong inverse sign (C1); M6 stale ℓ_4 hash; M7 the p10 orientation $+\partial_\mu B^\mu$ (C3 generic and conformal).

Polarization (one convention throughout). $V_4[h_1,h_2,h_3,h_4] := [t_1t_2t_3t_4]\,\ell_4(\sum_i t_i h_i) = D^4\ell_4(0)[h_1,h_2,h_3,h_4]$, the *unnormalised* fourth variation; the normalised symmetric form would be $V_4/4!$ and is not used. Where the factors enter D02: $1/4!$ appears only in $\ell_4(h) = \frac{1}{4!}V_4[h,h,h,h]$ and is cancelled by the 4! assignments of fields to legs; the vertex rule is $i\kappa^2 V_4[h_1,..,h_4]$ (i from e^{iS}, κ^2 from $S = \sum \kappa^{k-2}\int \ell_k$); the symmetry factor $\frac{1}{2}$ (two legs contracted) enters only at graph level. The same statement appears in `P13_quartic_verified.json`, `exports/V4_grav.json` and `exports/ell4_conventions.json`. `D02_vertex_available`=true, `D02_quantum_evaluated`=false.

23 BRST from finite geometric transformations

Builder/output: `brst_from_geometry.py`.

From $\delta g = \mathcal{L}_\xi g$ with $\xi = \kappa a c$ and the affine metric:

$$sh_{\mu\nu} = \tfrac{1}{\kappa}(\mathcal{L}_\xi g)_{\mu\nu} = a(\partial_\mu c_\nu + \partial_\nu c_\mu) + \kappa a(\mathcal{L}_c h)_{\mu\nu},$$
$$sc^\mu = \kappa a\, c^\rho\partial_\rho c^\mu, \qquad s\phi = \kappa a\, c^\rho\partial_\rho\phi, \qquad s\bar{c}_\mu = ib_\mu, \qquad sb_\mu = 0.$$

The ghost law $s\xi^\mu = \lambda\,\xi^\rho\partial_\rho\xi^\mu$ is an ansatz whose coefficient is *derived*: $s^2h = s^2\phi = 0$ on all jets (115 independent conditions) has the unique solution $\lambda = 1$. **Derived s_2 per sector:** $s_2 h = 0$, $s_2 c = 0$, $s_2\phi = 0$, $s_2\bar{c} = 0$, $s_2 b = 0$. Reason (a derivation, not the p09 policy "no invented s_2"): g is affine in h and $\mathcal{L}_\xi$ is linear in g, so $\kappa^{-1}\mathcal{L}_{\kappa ac}g$ has κ-degree exactly 1; $s\xi$ is quadratic in ξ, so sc has degree exactly 1; $s\phi$ is linear in ξ; the pair $(\bar{c}, b)$ is κ-independent. $s^2 = 0$ holds exactly to all orders. Discriminating control: for $g = \eta + \kappa h + \frac{\kappa^2}{2}h\eta^{-1}h$ the same construction gives $s_2 h \neq 0$, and dropping it breaks $s^2 = 0$ at $O(\kappa^2)$.

24 Source action

Builder/output: `source_action.py` (unchanged since p10).

$S_{\rm af} = \sum_A \sigma_A \int \Phi^*_A\, s\Phi^A$ (antifield on the left). Requiring the generator rule $(S, \Phi^A) = s\Phi^A$ with the XVI-N bracket *derives* $\sigma_A = (-1)^{|\Phi^A|}$: $\sigma_h = \sigma_\phi = +1$, $\sigma_c = \sigma_{\bar c} = -1$. The nonminimal coupling $-\int \bar c^{*\mu}\, ib_\mu$ is included; b^* does not occur ($sb = 0$). $X = S_{\rm af}$ has ghost number 0, is antifield-linear, and has 17 / 124 / 0 terms at $\kappa^0/\kappa^1/\kappa^2$. Because X is linear in antifields, $(S,S) = 2(S_{\rm field}, X) + (X,X)$ with $2(S_{\rm field}, X) = 2\int E_\Phi(S_{\rm field})\, \sigma s\Phi$ of antifield number 0 and $(X,X) = -2\sum \Phi^* s^2\Phi$ (mod d) of antifield number 1.

25 Gauge-fixing map in transformed variables

Builder/output: `gauge_fixing_map.py`.

$\Psi = -i\int \bar c_\mu (2D^\mu[h] - b^\mu)$. The orientation $\varepsilon = -1$ of $\Phi^*_A = \Phi^{*\prime}_A + \varepsilon\, \delta\Psi/\delta\Phi^A$ is derived from the Hessians, and $S_\Psi(\Phi, \Phi^{*\prime}) := S(\Phi, \Phi^{*\prime} - \delta\Psi/\delta\Phi) \simeq S + s\Psi$. **Hessians at generic momentum (since p11).** The (h, b) Hessian M is computed for a *symbolic* $k = (k_0, k_1, k_2, k_3)$. With $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, $\det D = -16 \neq 0$, and the Schur complement satisfies $A - BD^{-1}C = k^2 P$ identically in $\mathbb{Q}(k)$, with $\det(k^2 P) = 64(k^2)^{10} \neq 0$. The block-inverse formula then gives $(M^{-1})_{hh} = (k^2 P)^{-1}$ (graviton kernel $K = -\Box P$). The ghost kinetic term $i\bar c \Box c$ and the scalar $-\Box$ are exact as well. No momentum sample is used. **Canonicity** is checked on all 23 antifield components (276 antifield/antifield pairs and 23 antifield/field pairs). **Interface:** $\Gamma_\Psi(\Phi, \Phi^{*\prime}) := \Gamma(\Phi, \Phi^{*\prime} - \delta\Psi/\delta\Phi)$ is a *defined classical pullback/interface* along the canonical antifield shift. It is *not* an independently established quantum gauge-change theorem; no quantum Γ is constructed here.

26 Master equation through $O(\kappa^2)$ by antifield number

Builder/output: `master_kappa2.py`, `P13_master_kappa2.json`.

The κ^2 equation is $2(S_0, S_2) + (S_1, S_1) = 0$. The engine parts (antifield, ghost and scalar) are exact symbolic, as in p10. **Since p11,** the gravitational Noether rows are now *deterministic*: $W_n = s_0\ell_{n+2} + s_1\ell_{n+1}$ is built as an exact jet polynomial from the sha-checked exports, and $E_{c^\mu}(W_n) \equiv 0$ is proved for $n = 0, 1, 2$. The plane-wave evaluation is kept only as a regression control.

Lemma 3 (Multidegree, corrected)**.** *Let P_d project onto field multidegree d and let E_u be the Euler operator of field u. Then $E_u P_d = P_{d-e_u} E_u$. (E_u lowers the u-degree by one and preserves the others; total derivatives preserve multidegree.) Consequently, if $E_u F = 0$ for all u, then $E_u(P_d F) = P_{d-e_u}(E_u F) = 0$ for all d and u, so every homogeneous component $P_d F$ ($|d| \geq 1$) is a divergence; conversely a sum of divergences is a divergence.*

(p10 stated $E_u P_d = P_d E_u$, which is false; the conclusion survives with the corrected intertwining. It is checked on a test polynomial in `tests/test_p14.py`.) The divergence test is applied to the summed component; per-bracket-pair pieces are reported only to exhibit cancellations.

order	component	terms	divergence	per-pair (individually non-divergent)
κ^0	af0 grav Noether ($s_0\ell_2$)	exact	yes	—
	af0 (X_Ψ, X_Ψ) transformed	32	yes	$h, \bar c$
	af1, af2 (both variable sets)	0	yes	—
κ^1	af0 grav ($s_0\ell_3 + s_1\ell_2$)	2148	yes	s_0 alone: 159 per μ
	af0 scalar $2(L_0, X_1) + 2(L_1, X_0)$	20	yes	each piece
	af0 (X_Ψ, X_Ψ) transformed	372	yes	$h, c, \bar c$
	af1 invariant / transformed	180 / 180	yes	h, c
κ^2	af0 grav ($s_0\ell_4 + s_1\ell_3$, $s_2 = 0$)	15428	yes	s_0 alone: 2088 per μ
	af0 scalar $2(L_1, X_1) + 2(L_2, X_0)$	20	yes	each piece
	af0 (X_Ψ, X_Ψ) transformed	2392	yes	h, c
	af1 invariant / transformed	1356 / 1356	yes	h, c, ϕ
	af2 (both)	0	yes	—

Proposition 4 (Classical master equation of $S_{\rm inv}$ through $O(\kappa^2)$)**.** *Under the declared conventions every component of (S,S) through $O(\kappa^2)$ is a total divergence, in invariant and in Ψ-transformed variables, with all mixed cancellations retained:* `required_master_residuals=[]`.

Evidence (deterministic). Engine parts: exact Euler-operator divergence test. Gravitational af0: exact polynomial identity $E_{c^\mu}(W_n) \equiv 0$ on the exports. Structurally, $\sqrt{-g}R$ is a scalar density, so $\delta_\xi(\sqrt{-g}R) = \partial_\mu(\xi^\mu\sqrt{-g}R)$; $L_{\Gamma\Gamma} = \sqrt{-g}R + \partial_\mu B^\mu$ (C3, corrected orientation), and $s\,\partial_\mu B^\mu = \partial_\mu s B^\mu$. af1 rows are $-2\sum \Phi^* s^2\Phi$ and vanish by the derived nilpotency. □

The master certificate has no unconditional `pass` field. `execution_pass` (it ran), `claim_status` (`verified` only if there are no residuals, the hashes match and the quartic prerequisite is verified) and `scope` are separate.

27 *ch* dependency audit and the enumeration lemma

Builder/output: `ch_source_audit.py`.

Pattern: one c, one h; no $\phi, \bar{c}, b$ or antifields; ghost number 1; κ^2; one loop; off-shell. The p10 audit is unchanged. Six rows are possible and three survive; no coefficient is evaluated in this release: the h^*c row (D13); the $K'R_0(c)$ row (hh: D01, D02, D03, D11, D12; D11 is the inherited O5 kernel constructed under prescription, not re-evaluated in this release); and the order-κ one-point row (D14a–c, distinct from D12). The $\bar{c}^{*\prime}$, $b^{*\prime}$, ϕ and $(S_2,\cdot)$ rows are derived zero; `names_TBD`=[].

Lemma 5 (One-loop enumeration, `max_len`=2)**.** *(i) A connected one-loop graph is 1PI iff every internal line lies on its unique cycle, since a line off the cycle is a bridge. So after the quadratic (κ^0, two-field) terms are absorbed into propagators, a one-loop 1PI graph is a cycle of interaction vertices, each with exactly two internal legs. (ii) Every retained interaction vertex has coupling order κ^n with $n \geq 1$. Mechanically, the κ^0 inventory contains only two-field monomials. (iii) Hence a $\kappa^{\leq 2}$ contribution has at most two vertices, and* `max_len`*=2 is exhaustive. (iv) The auxiliary b occurs only in κ^0 bilinears (b^2, bh, $b\bar{c}^*$; checked). Its Gaussian elimination produces only bilinear (kinetic or source-bilinear) terms, never an omitted interaction. (v) Source-linear monomials (one quantum leg) cannot sit on the cycle. Attached off it, they create a bridge, i.e. a one-particle-reducible graph, so they cannot create extra 1PI coefficients.*

The relation $\Gamma_\Psi \leftrightarrow \Gamma$ used to read off rows is the defined classical pullback/interface above, not a quantum gauge-change theorem.

28 Acceptance propagation and release gate

Builder/output: `release_p14.py` (`classical_block`: verbatim fork of the p13 `release.py`, which is frozen and no longer run), `RELEASE_STATUS.json` (key `classical`), `P14_release.json`, `P14_closure_release.json`.

`release_claim_accepted` := `execution_pass` ∧ readiness ∧ (every deterministic prerequisite: C1–C7, quartic mutations rejected, deterministic grav Noether in the master, generic-k Hessian, full canonicity, enumeration lemma) ∧ hash consistency (ℓ_k export file = quartic certificate = master input). The closure artifacts are published only if `release_claim_accepted` and readiness both hold. Otherwise the release deletes any stale closure artifact, writes `P14_release_rejected.json` and exits nonzero, and the build stops before any PDF, zip or sync. End-to-end mutation tests (`tests/test_p14.py`, each in an isolated copy of the package) run mutation → stage residual or failed prerequisite → flag false → readiness false → publication rejected. They cover a flipped odd-derivative sign, a corrupted ℓ_4 coefficient in the export file, a dropped h^*c row, an altered gauge shift, an omitted c^* term, and a stale ℓ_4 hash (the recorded hash is actually mutated).

29 Inherited D11 context (o70-sync, read-only)

The following inherited notes are reproduced unchanged for context; they are not modified by p13 or p14. **o70-sync.** Four mathematical prerequisites Constructed: fixed u_μ^R (analytic-family residue $r = -i\pi^2$, proof `lem:analytic_FP_u_mu_R_o70`); global $T = \mathcal{K}u$ with α, β *derived* ($A^2/60$, $A^2/18$), all 256 components symbolic; W_w jet-pullback as the *only* active reference (`W_w_jet_pullback_via_Pi_tr_o68`); contact moments for L with `kernel_two_point_Ward` (not functional ST4b). Kernel counterterm earned under prescription after prerequisites pass. UV subtraction vs finite repair kept separate. Corpus $\mathcal{P}$ (trace reversal) not reused for $P_{\rm tr}$. Bulk `symmetry_restored=false`; O.1 NOT_CLOSED. $\Pi^R_{\rm tr}|_U = (i\kappa^2/4)T$ (prefactor retained). $s_0 C_L$ not evaluated (minimal release).

30 Massless scalar matter-loop subsector (o70-sync)

Subsector verdict: Constructed under prescription (kernel).

Four prerequisites Constructed; Gate 3 spanning still Partial (candidate_list). Bulk O.1 NOT_CLOSED; insertion freeze unchanged; Em01 not a premise. $D11_kernel_counterterm$ = earned only after math prerequisites pass (development mark: Conditional).

30.1 Gate 1: mixed scalar Noether (retained)

Coupling-stripped $S_{\phi,i} = \frac{1}{2}\int a_i^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ ($a_0 = \eta$). With BRST ghost normalization $a = \frac{1}{2}$ (distinct from D11 coefficients α, β) and $sq = \kappa a\,\mathcal{L}_c q$, the densitized identities

$$s_0 a_1 = a\,\mathcal{D}_c\eta, \qquad s_0 a_2 + s_1 a_1 = a\,\mathcal{D}_c a_1$$

are verified by jet-polynomial substitution (executable). The boxed density identity

$$\boxed{s_0\mathcal{L}_2 + s_1\mathcal{L}_1 = a\,\partial_\rho(c^\rho\mathcal{L}_1)}$$

follows and integrates under compact support. This is the mixed metric–scalar Noether cancellation (Gate 1), not an independent scalar BV master equation. Separate: (i) invariant baseline $S_{\rm inv}$; (ii) prepared-action defect (S_D, S_D); (iii) insertion-class obstruction (XVI-N thm:massless-obstruction).

30.2 Gate 2: normalized D11 — prefactor Constructed; tensor Constructed

Route A (Wick) independently yields

$$\log Z\big|_{\rm bubble} = -\frac{\kappa^2}{8}\int h\,T\,h, \qquad \Gamma_{1,\rm matter} = -i\log Z, \qquad \Pi_{\rm bubble} = \frac{i\kappa^2}{4}\,VV(CC + CC).$$

Route B expands $\log Z = -\frac{1}{2}\operatorname{Tr}\log H$ to $(-\kappa^2/4)\operatorname{Tr}(D_F H_1 D_F H_1)$. Symmetry of a_1 gives $\mathrm{comb}_B = \frac{1}{2}$ (Constructed — not unresolved) and $\Pi_B = i\kappa^2/4\,T$. Tensor witness $T_{0000} = 24A^2/r^8$ at $z = (0, r, 0, 0)$.

30.3 Gate 3 / Deliverables 1–3

Scalar $u_\mu^R = \mathrm{FP}_{\zeta=0}[\mu^{2\zeta}(Q - i0)^{-2+\zeta}]$ with $r = -i\pi^2$ from analytic-family recurrence + Gate 0 ($\Box(Q - i0)^{-1} = 4\pi^2 i\delta$), not Wick-Jacobian alone. Global $T = \mathcal{K}u$ on U ($\mathcal{K} = \alpha\mathcal{D}^{(2)} + \beta\mathcal{D}^{(0)}$; α, β derived; all 256 components; lemma `lem:BV_rules_on_U`). Active W_w reference only: $\langle\Pi^R_w, f\rangle = \langle\Pi^R_{\rm tr}, W_w f\rangle$ (prescription `W_w_jet_pullback_via_Pi_tr_o68`). Old cutoff ARCHIVED; equivalence NOT claimed. False pointwise $|T| \le C|z|_E^{-8}$ hypothesis removed (annular seminorms). Spanning set still `candidate_list` ⇒ Gate 3 Partial overall.

30.4 Gate 4 / Deliverable 4: contact moments

$$M_\gamma = \langle \Pi_{\mathrm{tr}}, w z^\gamma/\gamma! \rangle, \qquad L = \sum_{|\gamma|\le 4} (-1)^{|\gamma|} M_\gamma \, \partial^\gamma \delta.$$

Odd moments vanish by evenness. $\Pi_w + L = \Pi_{\mathrm{tr}}$; $\partial(\Pi_w + L) = 0$ both pairs (`kernel_two_point_Ward`; not functional ST4b). L order ≤ 4; kernel-divergence contact bound on ∂L is five. Formal transpose (not L^2 adjoint). L not claimed Lorentz covariant. $C_L = \frac{1}{2}\int h\, L\, h$ (gh= 0, even); $s_0 C_L$ not evaluated. Kernel CT $\neq$ bulk BV restore.

31 Fixed scalar extension u_μ^R (o70-sync)

Prescription. One analytic finite-part scheme:

$$u_\zeta(z) = \mu^{2\zeta}(Q - i0)^{-2+\zeta}, \qquad u_\mu^R = \mathrm{FP}_{\zeta=0}\, u_\zeta.$$

Frozen: $\eta = \mathrm{diag}(+---)$; $Q - i0$; principal logarithm; $\mu > 0$ (inverse length); finite part = Laurent constant term; **no** additional finite $c\delta$. Proof identifier: `lem:analytic_FP_u_mu_R_o70` (equivalent FP prescription to o68/o69; constructive holomorphy proof added).

Lemma (distributional analytic continuation).

1. u_ζ admits a distributional meromorphic continuation near $\zeta = 0$ against arbitrary compact smooth tests, with a simple pole.
2. Holomorphy on tests supported in U: the pairing equals $\langle (Q - i0)^{-2}, f\rangle$ at $\zeta = 0$.
3. The pole part is local: $\dfrac{r\, f(0)}{\zeta}$ with $r = -i\pi^2$, derived below — not by naming analytic regularization.
4. u_μ^R is Lorentz covariant and even under $z \mapsto -z$.
5. $\mathrm{sd}(u_\mu^R) \le 4$; residual $c\delta$ freedom frozen by excluding any added contact (XVI-J `prop:extension` = freedom only).

Ordinary integrability near the cone. Near a nonzero null point with local coordinate $t = Q$, ordinary absolute local integrability of $(Q - i0)^{-2+\zeta}$ requires $\int_{-\varepsilon}^{\varepsilon} |t|^{-2+\Re\zeta}\, dt < \infty$, hence $\Re\zeta > 1$. At $\zeta = 0$ the U-supported pairing is obtained by boundary-value / analytic-family continuation, not by an ordinary integral.

Proof that $\phi_{-1+\zeta}$ is holomorphic near $\zeta = 0$. Do *not* argue from the zero of $4\zeta(\zeta - 1)$ alone.

1. Let $F(\zeta) = (Q - i0)^\zeta$. This is a holomorphic family of distributions near $\zeta = 0$: instantiate Dütsch–Fredenhagen–Keller–Rejzner, arXiv:1311.5424, Definition 4.2 (analytic regularization) and §4.1–4.2 (complex powers / meromorphic two-point families), with hypotheses $Q - i0$, principal logarithm, tests in $\mathcal{D}(\mathbb{R}^4)$, and Gate 0 normalization. (Classical support: Gelfand–Shilov / Hörmander complex powers $(x \pm i0)^\lambda$ are entire in λ.)
2. Const-coeff differentiation under the pairing gives the recurrence $\Box\phi_\lambda = 4\lambda(\lambda + 1)\phi_{\lambda-1}$, hence
$$\Box F(\zeta) = 4\zeta(\zeta + 1)\, \phi_{-1+\zeta}.$$
3. $F(0) = (Q - i0)^0 = 1$, so $\Box F(0) = 0$. Holomorphy of F yields $\Box F(\zeta) = \zeta\, G(\zeta)$ with G holomorphic in $\mathcal{D}'$.
4. Therefore
$$\phi_{-1+\zeta} = \frac{\Box F(\zeta)}{4\zeta(\zeta + 1)} = \frac{G(\zeta)}{4(\zeta + 1)}$$
is holomorphic near $\zeta = 0$.

Residue (only after the holomorphy premise). Gate 0 gives $D_F = -A(Q - i0)^{-1}$ with $A = 1/(4\pi^2)$ and $(-\Box)D_F = i\delta$, hence

$$\Box(Q - i0)^{-1} = 4\pi^2 i\,\delta.$$

The second recurrence

$$\Box\phi_{-1+\zeta} = 4\zeta(-1+\zeta)\,\phi_{-2+\zeta}$$

and the holomorphy of $\phi_{-1+\zeta}$ yield

$$\langle \phi_{-2+\zeta}, f\rangle = \frac{r\,f(0)}{\zeta} + \mathrm{FP} + O(\zeta), \qquad \boxed{r = -i\pi^2}.$$

Scale $\mu^{2\zeta}$ affects only the finite part. Operational Laurent identity:

$$\langle u_\zeta, f\rangle = \frac{r\,f(0)}{\zeta} + \langle u_\mu^R, f\rangle + O(\zeta).$$

Euclidean radial integral ($|S^3| = 2\pi^2$) is an *independent sign check*, not the sole justification. Not momentum-space MS; not XVI-J's value-jet extension of $d(z)^2$.

32 Track II: insertion status freeze

Retained objects. Seed $F_t = \langle t_D, h \otimes h \otimes \phi\rangle$ (XVI-I; no extra factorial) and the regular, seed-preserving, antifield-linear class $I_t = F_t + A_0 + \kappa A_1$ with $A_0 = \int(h^*U + \phi^*V + c^*W)$.

Frozen negative / conditional result. Under XVI-N massless-obstruction hypotheses, on the constructed massless special-family witness: no such I_t satisfies $Q_{\mathrm{inv}} I_t = O(\kappa^2)$. Route A is sufficient and discharged; Route B is not claimed discharged. Certified small-mass interval $0 \le m < 1.3715 \times 10^{-4}$ at frozen geometry; arbitrary prescribed $m > 0$ unresolved.

Hard rule. No quantum completion is claimed whose classical leading term contradicts this obstruction. Extra antifields or nonminimal fields do not automatically remove the order-κ^0 field-only contradiction ($P_0 Q_0 A_0 = 0$ on the free shell).

O.9 separation. Existence of a renormalized product of the bare F_t is not closure of F_t in the BV/BRST sense. N-Ins.5 remains structural only after invariant completion. Success on the bulk quantum sector (verdict 1) does not complete F_t (verdict 3).